DESIGN, IMPLEMENTATION, AND ANALYSIS OF A LOW-POWER SUCCESSIVE APPROXIMATION REGISTER ANALOG-TO-DIGITAL CONVERTER FOR BIOMEDICAL APPLICATIONS

A THESIS

Presented to the Department of Electrical Engineering

California State University, Long Beach

In Partial Fulfillment

of the Requirements for the Degree

Master of Science in Electrical Engineering

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May 2025

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ABSTRACT

The demand for wearable, battery-operated devices capable of handling biological signals is rising due to the rapid advancements in healthcare systems. Typically, these signals are analog and require conversion to digital format for additional processing and analysis. Successive Approximation Register Analog-to-Digital Converters (SAR ADC) are optimal for low-power applications because of their energy-efficient design, making them ideal for such systems. The key components of a SAR ADC include the Sample-and-Hold (S/H) circuit, the Comparator, the Digital-to-Analog Converter (DAC), and the SAR control logic. This work presents an innovative design methodology, which incorporates a low-power S/H circuit, an enhanced SAR logic structure (introducing a modified D logic that reduces the transistor count in the SAR logic, leading to significant power savings), and an optimized DAC to enhance power efficiency and signal integrity. The proposed 10-bit SAR ADC is implemented using a $45nm$ CMOS process in Cadence Virtuoso, operating at a supply voltage of $1V$. The ADC is optimized for biomedical applications, prioritizing low power consumption and precise signal processing. This study introduces a notable enhancement in the design of SAR ADCs, reducing power consumption and enhancing signal precision, making it well-suited for next-generation biomedical systems.

Keywords: CMOS, Sample and Hold, Comparator, Digital-to-Analog Converter, SAR Logic, SAR ADC.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to Dr. Ava Hedayatipour for her exceptional guidance and mentorship throughout my academic journey. Her expertise, constructive feedback, and consistent encouragement have been pivotal to the successful completion of this work. I am deeply thankful for her dedication and commitment, which has significantly enhanced the quality of my research and helped me navigate the challenges of this project.

I would also like to express my sincere gratitude to my thesis committee members, Dr. Duc Tran and Dr. Sean Kwon, for their willingness to serve on my committee. It has been an honor to present my research to such esteemed members of the CSULB faculty, and I deeply value this opportunity.

Furthermore, I would like to extend my sincere thanks to Dr. Ebrahim Amiri and the instructors in the Department of Electrical Engineering for their support and guidance during my time at CSULB. Their expertise and direction have been crucial to my academic development and the successful completion of my research.

Lastly, I am deeply grateful to my parents, Anand Kumar Talluri and Surekha Kayatham, for their constant love, patience, and belief in my abilities. Their unwavering support has been a source of strength and inspiration, and I am truly fortunate to have them by my side.

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CHAPTER 1

INTRODUCTION

The efficiency of data transmission has become a critical factor in modern communication systems. Analog signals, which form the foundation of many of the technologies we interact with daily—such as televisions, radios, and smartphones—have the unique ability to convey an infinite range of data. This quality allows for easy processing and virtually endless possibilities for usage, contributing to the widespread development of devices that enhance convenience and improve the quality of life.

In various industries, particularly in consumer electronics, healthcare, and automotive sectors, sensors have become integral components. These sensors monitor and measure physical properties, producing electrical signals in response to external stimuli. Many different applications use sensors, including touch-based interfaces in contemporary electronics and temperature-regulating systems. To solve environmental problems and maximize resource utilization, sensors are also essential [1]. The growing use of wireless and internet-connected systems, especially those based on System-on-Chip (SoC) solutions, shows how important it is to convert and send analog data correctly in a world that is becoming more and more digital.

Converting continuous analog signals into a digital format that computers and other digital systems can process is one of the difficulties engineers have in this digital field. ADCs are made especially for this use. They convert continuous analog input signals, such as voltage or current, into binary outputs that digital systems can process. ADCs are essential for bridging the gap between the analog and digital worlds because a large portion of the data processed in modern electronics is in analog form [1].

ADCs manage mixed-signal and radio-frequency activities, as well as analog and digital inputs, in SoC architectures. Systems may now perform more complex tasks, such voice analysis, with reduced power consumption because of advancements in integrated circuit (IC) technology. In biomedical electronics, where energy limits are a significant concern, this capacity is particularly

crucial. The need for energy-efficient ADCs has never been higher due to the rising need for portable, low-power biomedical equipment, such as wearable sensors and implantable medical devices. These ADCs can successfully lower overall power consumption by employing low-power complementary metal-oxide semiconductor (CMOS) design approaches, which is crucial for biomedical equipment.

The main goal of this work is to develop a power-efficient type of ADC using specially developed $45nm$ manufacturing technology. It was specifically designed to be used in biological applications. The goal is to build an ADC that meets the size and performance requirements for equipment used for numerous mobile medical applications and ongoing patient observation while using less power.

1.1 Need for SAR ADCs in Biomedical Applications

Medicine and healthcare systems have advanced significantly as a result of the increased need for efficient disease diagnosis and treatment. Consequently, biomedical advancements such as wireless sensors and medical implants have emerged as crucial tools for continuous patient monitoring enabling a seamless return to everyday life without compromising patient comfort [2].

An important component driving these developments is the ADC, which is essential for converting analog biological signals into digital data for processing, analysis, and storage. Because portable devices have a limited battery life, reducing power consumption has become one of the main difficulties in the context of wireless biomedical applications [3]. The ADC, which acts as a bridge between the analog and digital processing domains, is essential to ensuring the efficiency and use of these devices [2]. These ADCs need to have moderate resolution and sample speeds—typically between a few KS/s and a few MS/s— and low power consumption to extend battery life [4], [5]. Additionally, ADCs must be small to fit inside the limitations of tiny bioelectronic devices [6].

ADCs are characterized by several performance metrics, including sampling frequency (FS), resolution, effective number of bits (ENOB), signal-to-noise-and-distortion ratio (SNDR), power

consumption, figure of merit (FOM), differential non-linearity (DNL) and integral non-linearity (INL). The simultaneous need for high resolution and speed presents a design challenge, especially in low-power applications [7]. While significant progress has been made in ADC design and implementation, the challenge of achieving both high speed and high resolution with minimal power consumption remains critical [8].

In response to these challenges, the SAR ADC has emerged as a promising solution for wireless biomedical applications [9]. It provides an effective way to transform analog signals into digital data while preserving low hardware complexity because of its straightforward architecture and low power consumption. This advantage qualifies the SAR ADC for energy-efficient biomedical systems, which are necessary to satisfy the requirements of contemporary medical technology.

1.2 Advantages of SAR ADCs for Low-Power Biomedical Applications

For biomedical applications, particularly in implantable or portable devices, minimizing power consumption is essential. The SAR ADC has emerged as a top option among ADC architectures because of its capacity to provide excellent resolution with low power consumption. Its design relies on a limited number of analog components, typically just a single comparator, resulting in a compact and straightforward implementation.

TABLE 1. Advantages and Disadvantages of Different ADC Architectures

Type of ADCs	Advantages	Disadvantages
FLASH	Very Fast	Needs many parts (255 comparators for 8-bit ADC), Lower resolution, Expensive, Large power consumption
Sigma-Delta	High Resolution, No precision external components needed	Slow due to oversampling
Pipeline	High throughput rate, Low power consumption	Slow, Requires minimum clock frequency
Successive Approximation	Capable of high speed, Medium accuracy compared to other ADC types, Good tradeoff between speed and cost	higher resolution successive approximation ADCs will be slower

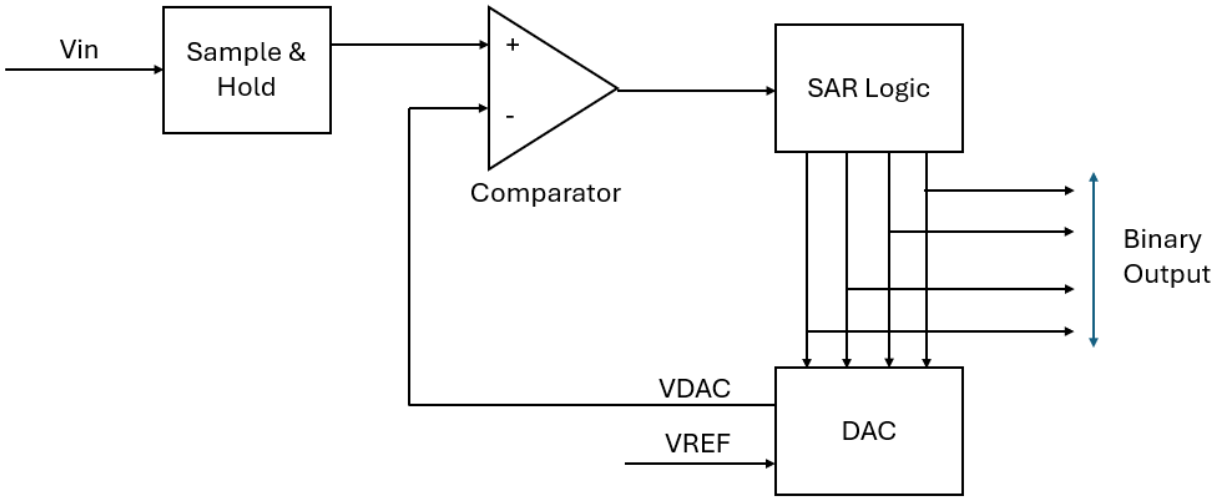


FIGURE 1. Block diagram of SAR ADC.

This makes the SAR ADC particularly well-suited for applications requiring moderate speed and resolution. In addition, the SAR ADC is very flexible and can work with bioelectronic devices because most of its parts are digital (except for the comparator) [10]. Table 1 highlights why SAR ADCs are the recommended option for battery-operated biomedical equipment by summarizing the benefits and drawbacks of different ADC topologies [1].

1.3 Operation of Successive Approximation ADC

A conventional SAR ADC consists of a S/H, a dynamic voltage comparator, SAR logic control, and a DAC, as shown in Fig. 1. The S/H circuit captures and maintains the analog input voltage V_{in}. The output voltage from the S/H circuit, $V_{S/H}$, is compared to the output voltage generated by the DAC, V_{DAC}, using the comparator. The outcome of this comparison, whether the sampled input is higher or lower than the DAC's output, determines the digital output, which can be either a high or low logic level. The SAR ADC performs N comparisons prior to generating a digital value that corresponds to the result of the N-bit binary comparison. The SAR ADC employs a binary search method to transform the input analog waveform into a discrete digital representation prior to finalizing the digital output for the current input.

1.4 Thesis Outline

Chapter 2 of this thesis report is a Literature Review. It discusses the basics of ADC, the most important performance metrics for ADCs, current SAR ADC architectures, and the difficulties in creating low-power SAR ADCs. It also highlights relevant advances in the field.

Chapter 3 introduces the proposed SAR ADC architecture focusing on the modified block implementation. It describes the design process used to maximize the power efficiency of several parts, including the DAC, SAR logic, and S/H circuit.

Chapter 4 presents the simulation results and comparisons of the proposed design, including performance evaluations based on resolution, speed, and power usage. We contrast the simulation results with those of current SAR ADC systems to demonstrate improvements.

Chapter 5 covers the design process of layouts, including design rule check (DRC), layout versus schematic (LVS), parasitic extraction, and post-layout simulation. It provides information on the verification of DRC, the analysis of LVS, and the impact of the layout on performance metrics.

Finally, Chapter 6 has a summary of the findings that highlights significant contributions, identifies research limits, and recommends areas for further study and advancement.

CHAPTER 2

LITERATURE REVIEW

This section examines the present state of research and developments in SAR ADCs, focusing specifically on their applications in biomedical s ystems. This review explores various SAR ADC architectures, emphasizing their advantages, drawbacks, and the progression of design methodologies aimed at enhancing performance, power efficiency, and integration in medical devices. This section aims to identify deficiencies in current technologies by examining the existing literature, thereby establishing a basis for the proposed solutions and illustrating how advancements in SAR ADCs can advance biomedical engineering and improve the effectiveness of medical devices.

In recent years, the design of SAR ADCs has gained attention due to their inherent advantages of low power consumption and moderate resolution. A notable example of this is the work by Naveen and Sonoli [11] in which they proposed a 10-bit SAR ADC designed specifically for biomedical applications. In their paper, the authors explore a low-power solution that operates with a single supply voltage of $1V$, offering an energy-efficient option for portable or implantable biomedical devices.

The paper introduces a 10-bit low-power SAR ADC simulated with $180nm$ CMOS technology. The literature indicates that utilizing a capacitive DAC shown in Fig. 2, results in reduced power usage compared to a resistive DAC. The capacitive DAC incorporates a S/H circuit.

A dynamic latch comparator shown in Fig. 3 is employed instead of an open-loop comparator to improve speed and decrease power consumption. This design decision enhances both speed and decreases power usage. Furthermore, transistors are miniaturized by diverse methodologies to attain optimal performance.

The SAR ADC is selected for its uncomplicated design methodology, as referenced in prior studies [12]. The SAR architecture shown in Fig. 4 integrates a ring counter and a code register for the purpose of storing the output bits. Rossi and Fucili's non-redundant SAR architecture [13]

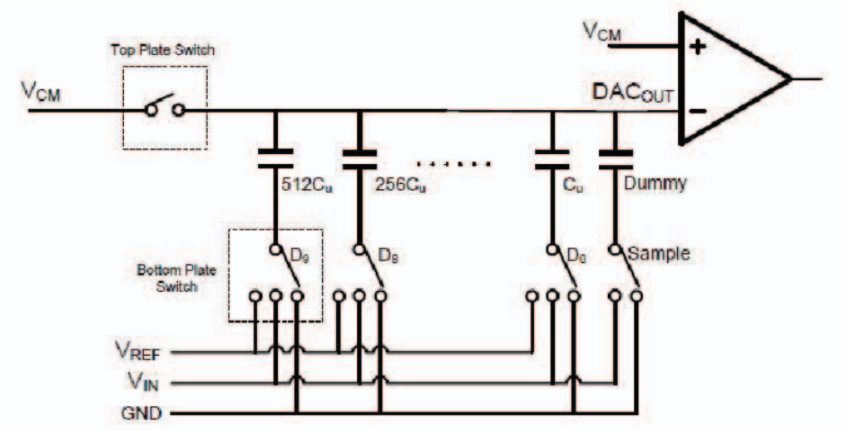


FIGURE 2. Proposed DAC schematic [11].

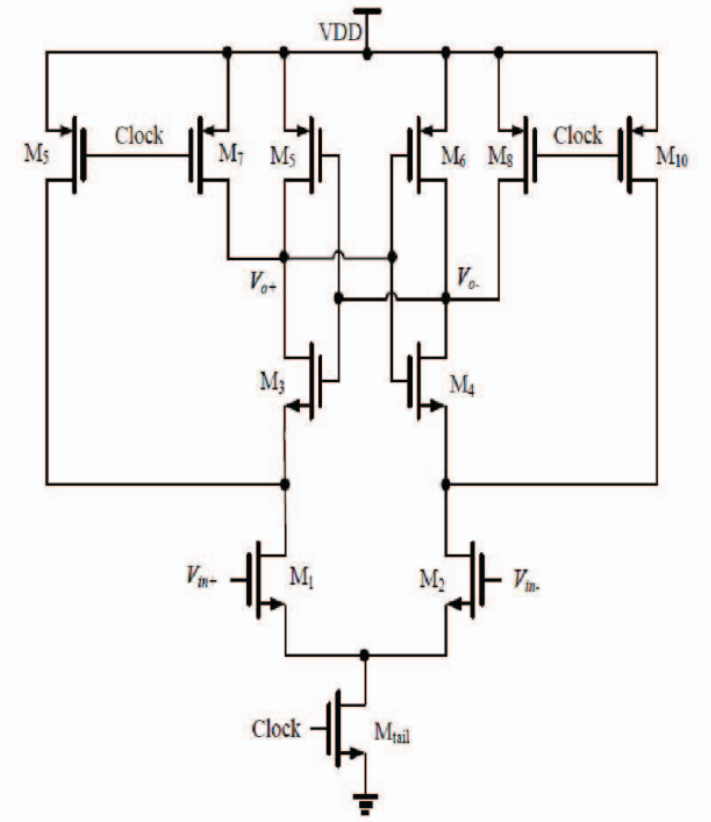


FIGURE 3. Dynamic latch comparator [11].

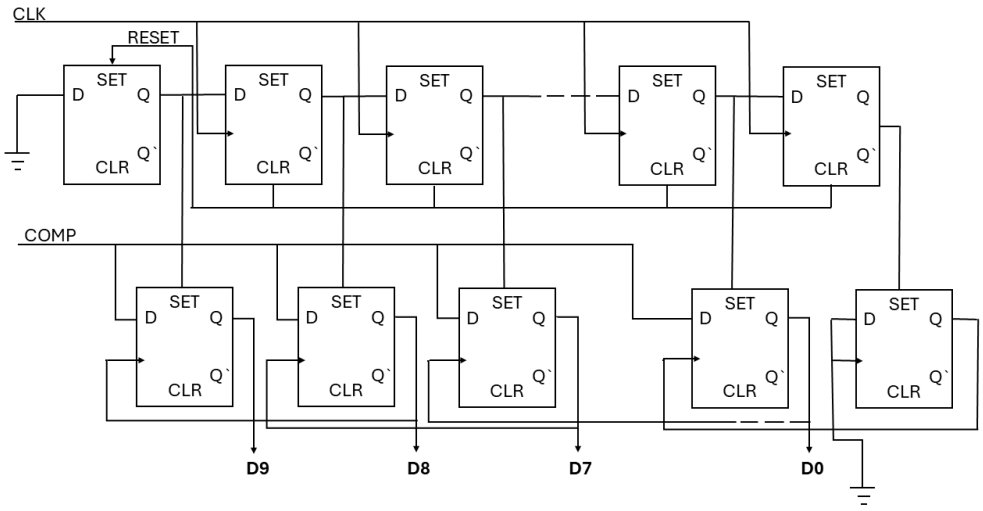


FIGURE 4. Successive approximation register [11].

utilizes fewer D Flip-Flops (DFF) than our approach, yet necessitates greater power consumption because of the additional combinational logic involved in its architecture.

The work [11] highlights the potential of SAR ADCs in enhancing biomedical electronics by providing low power consumption, compact size, and adequate resolution for many medical applications. The shift towards energy-efficient and compact ADC designs is essential due to the increasing demand for smaller, battery-powered biomedical equipment.

A study presented in [14] proposes an 8-bit SAR ADC intended for biomedical applications. The SAR ADC operates at a sampling rate of $10KS/s$ and requires $400nW$ from a $1V$ power source. The circuit is simulated with the Cadence Virtuoso tool utilizing $90nm$ CMOS technology, indicating that the suggested SAR ADC is a viable candidate for biomedical applications.

The ADC's performance during measurement is affected by modifications to the reference voltage. In the construction of the binary-weighted capacitor array for the DAC, the delay constant is considered to guarantee that switches managing varying capacitor values are proportionately constructed. The capacitor unit is sufficiently big to attenuate the kT/C noise, where k represents the Boltzmann constant, T denotes the temperature in Kelvin, and C signifies the total capacitance of

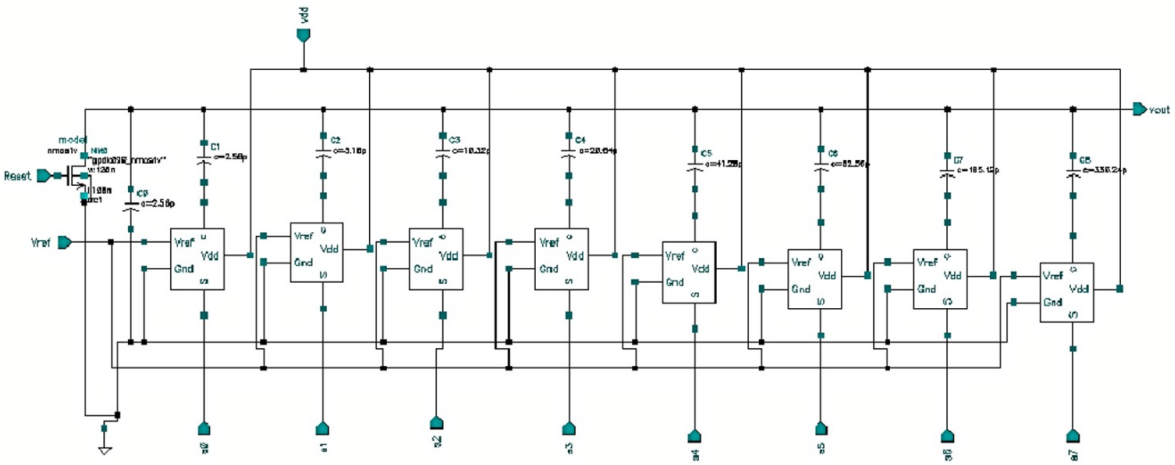


FIGURE 5. Schematic of 8-bit DAC [14].

the capacitor array in Farads. The substantial total capacitance effectively mitigates kT/C noise, making the resolution of the capacitor array mismatch a crucial concern.

The fundamental S/H circuit, referred to as the track and hold circuit, is a single-ended configuration including an NMOS switch and a sampling capacitor C_S. In this fundamental S/H circuit, charge injection and clock feed-through are critical concerns that emerge when a transistor functions as a switch in a switched capacitor circuit. An S/H circuit utilizing a dummy switch is a prevalent technique for mitigating charge injection.

There are two primary categories of comparators: static and dynamic. Static comparators, employed in medium-speed applications, maintain a continuous low-resistance connection from each gate output to either V_{DD} or V_{SS}, except during switching transients. Conversely, dynamic comparators momentarily retain signal values on the capacitance of high-impedance circuit nodes, rendering them appropriate for high-speed applications. Nevertheless, they exhibit heightened sensitivity to noise. The pursuit of increased speed and reduced power consumption in medium-to-high-speed applications renders dynamic comparators especially appealing, since they are energy efficient due to the lack of static current consumption, in contrast to static comparators.

The principal objective of this study is to minimize power consumption, rendering the

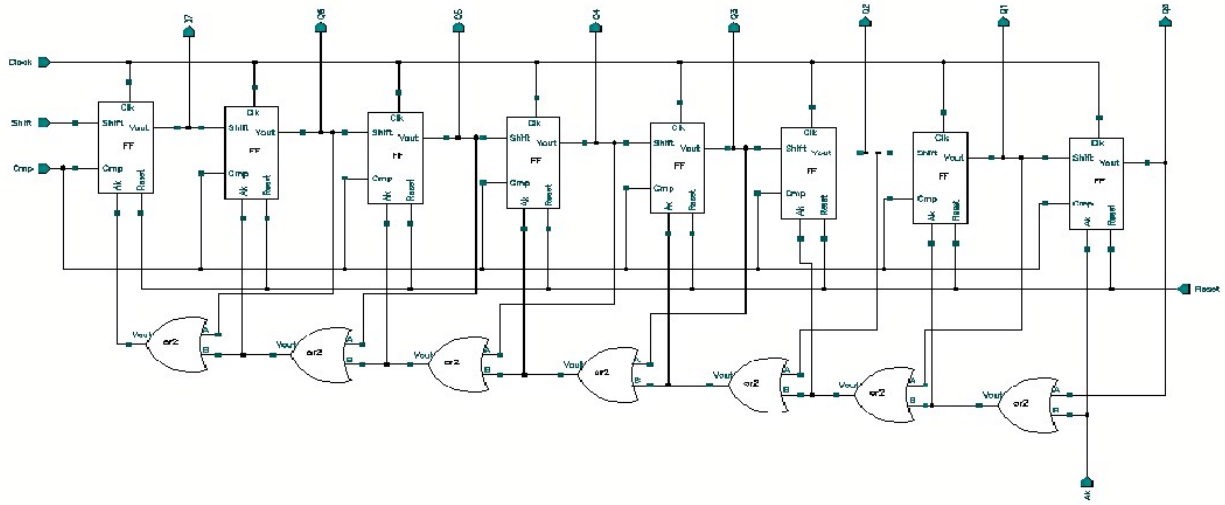


FIGURE 6. Schematic of the SAR logic [14].

binary-weighted DAC the most appropriate choice for low power usage. The binary-weighted DAC can be fabricated with arrays of binary currents (via resistor or current methodologies) or binary-weighted charge arrays. The latter is favored for the subsequent reasons: MOS devices utilized as charge switches exhibit zero offset voltage and possess very high input resistance when functioning as amplifiers. Moreover, capacitors can be readily manufactured via metal gate technology. Consequently, employing capacitors in lieu of resistors as precise components and utilizing charge instead of current as the operational medium markedly decreases power dissipation.

The schematic of the SAR ADC's control logic comprises the SAR unit and a series of OR gates shown in Fig. 6. The SAR unit is constructed utilizing a 3:1 multiplexer (MUX) and a DFF. When the OR input is elevated, the ADC enters a suspended state in which no conversions take place. Conversion recommences when the OR input is diminished. The circuit is evaluated under the presumption that the comparator output is '0'.

The single-ended DAC-based SAR ADC shown in Fig. 7 is designed utilizing 90nm technology in Cadence Virtuoso, functioning at a 1V supply voltage and a sampling rate of 10 KS/s. The circuits are simulated using an input signal characterized by a peak-to-peak voltage of 500mV and a frequency of 250Hz. The DNL and INL for the 8-bit SAR ADC are recorded as 0.2/-0.6 and

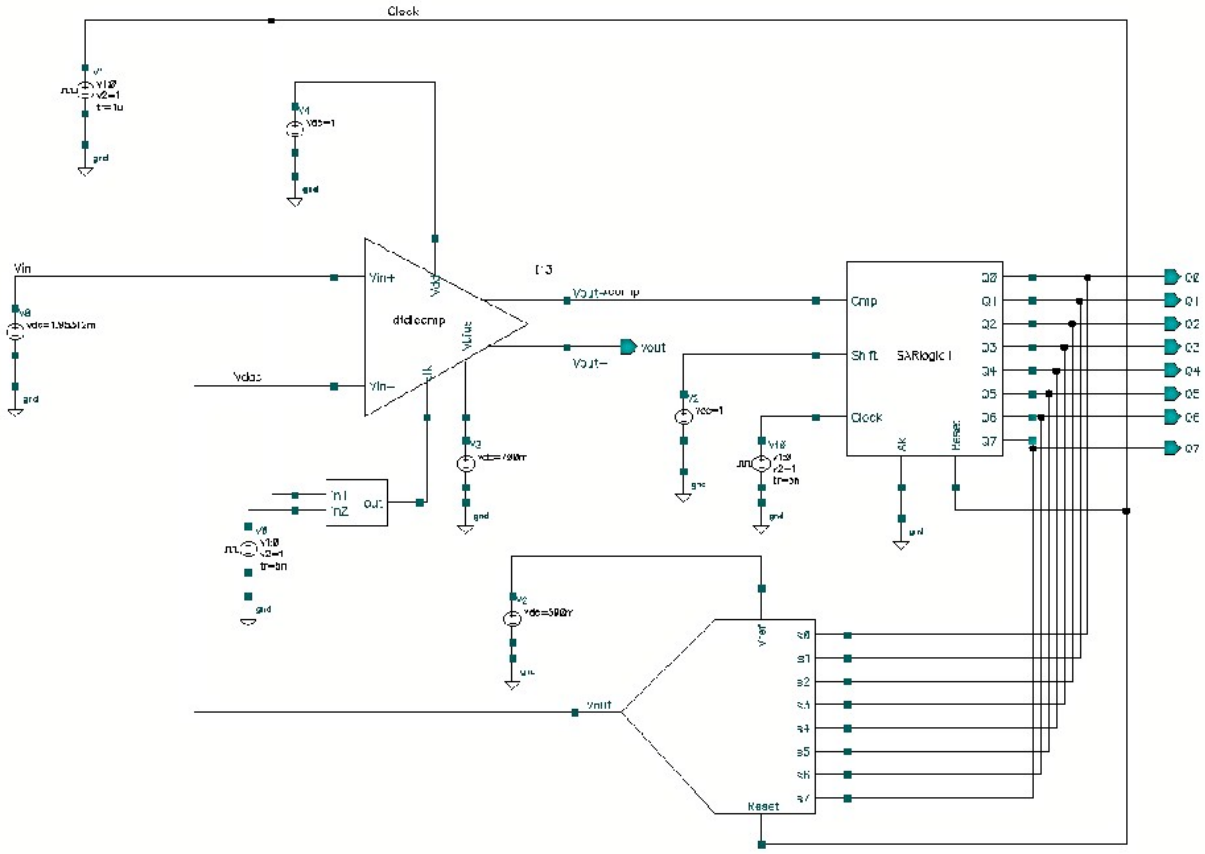


FIGURE 7. Schematic of 8-bit SAR ADC [14].

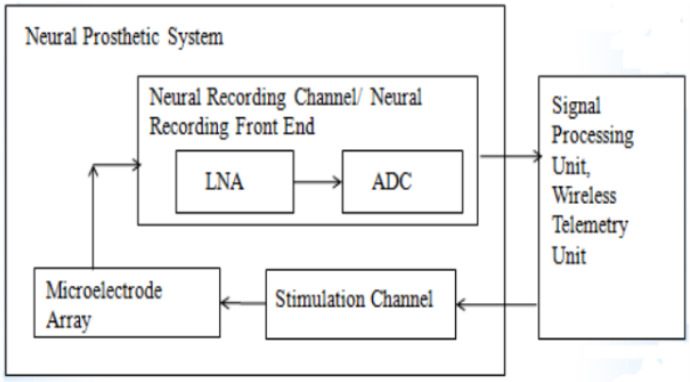


FIGURE 8. Implantable system for brain machine interface system [15].

0.46/-0.21, respectively. The SAR ADC consumes $400nW$ of power at a supply voltage of $1V$.

In their work, Bire and Gugulothu [15] present a novel approach for designing a 4-bit SAR ADC for neural prosthetic systems, particularly brain-machine interfaces (BMIs).

ADC are crucial elements in neural recording systems, especially within neural prosthesis systems such as BMI. These systems necessitate effective ADC architectures shown in Fig. 8, to transform analog signals from neural activity into digital format for processing. The SAR ADC shown in Fig. 9 is prevalent in low-power applications owing to its high precision and very straightforward construction. In recent years, energy economy in SAR ADC designs has emerged as a crucial consideration, particularly for applications like brain device, where battery longevity and energy consumption are essential.

A primary aspect influencing the energy consumption of SAR ADCs is the DAC employed to produce reference levels during the conversion process. Multiple DAC designs can be utilized in SAR ADCs, with the capacitive DAC (C-DAC) being the predominant choice due to its benefit of negligible static power dissipation. The C-DAC functions by charging and discharging capacitors to generate the necessary reference voltages, which is a more efficient method than resistor-based DACs. Nonetheless, the energy consumption in the DAC remains substantial, and mitigating this energy usage is a focal point of ongoing research.

Bire and Gugulothu [15] focus on the quaternary capacitive DAC switching strategy (QCS)

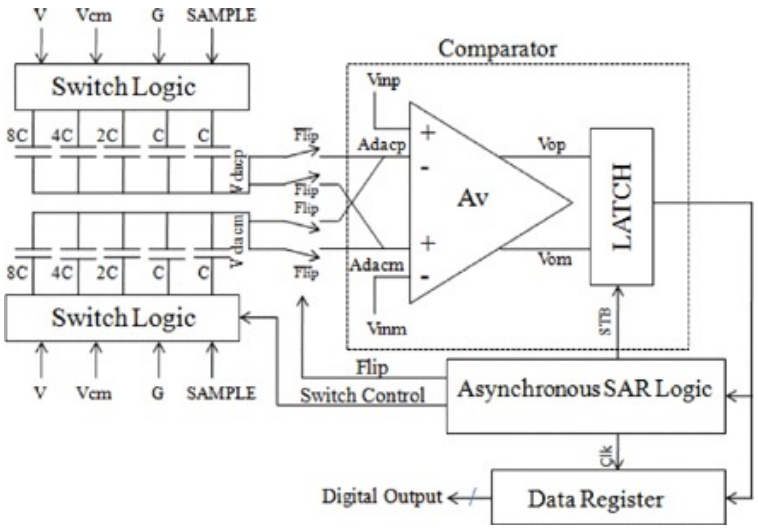


FIGURE 9. Proposed architecture of 4-bit SAR ADC [15].

shown in Fig. 10, which minimizes energy consumption in DAC by minimizing switching capacitance. In the proposed QCS [15], the most significant bit (MSB) capacitors are excluded from the estimation of the least significant bits (LSBs), thereby substantially decreasing the energy consumption from the reference.

The main benefit of the QCS scheme is that it renders energy consumption in the C-DAC independent of the output digital code, in contrast to traditional methods where energy consumption is contingent upon the created digital code. The energy independence enhances the SAR ADC's efficiency by lowering the total energy expenditure during the conversion process. The proposed technology realizes a 50% decrease in average energy usage relative to traditional switching methods.

Recent studies by Bire and Gugulothu [15] on low-power ADCs emphasize the need of employing modern manufacturing methods to enhance energy efficiency. The deployment of SAR ADCs on 0.25μm CMOS technology demonstrates an effective equilibrium between power consumption and performance. Their study presents a SAR ADC developed with 0.25μm CMOS technology, thereby improving its energy efficiency and rendering it appropriate for low-power brain recording systems.

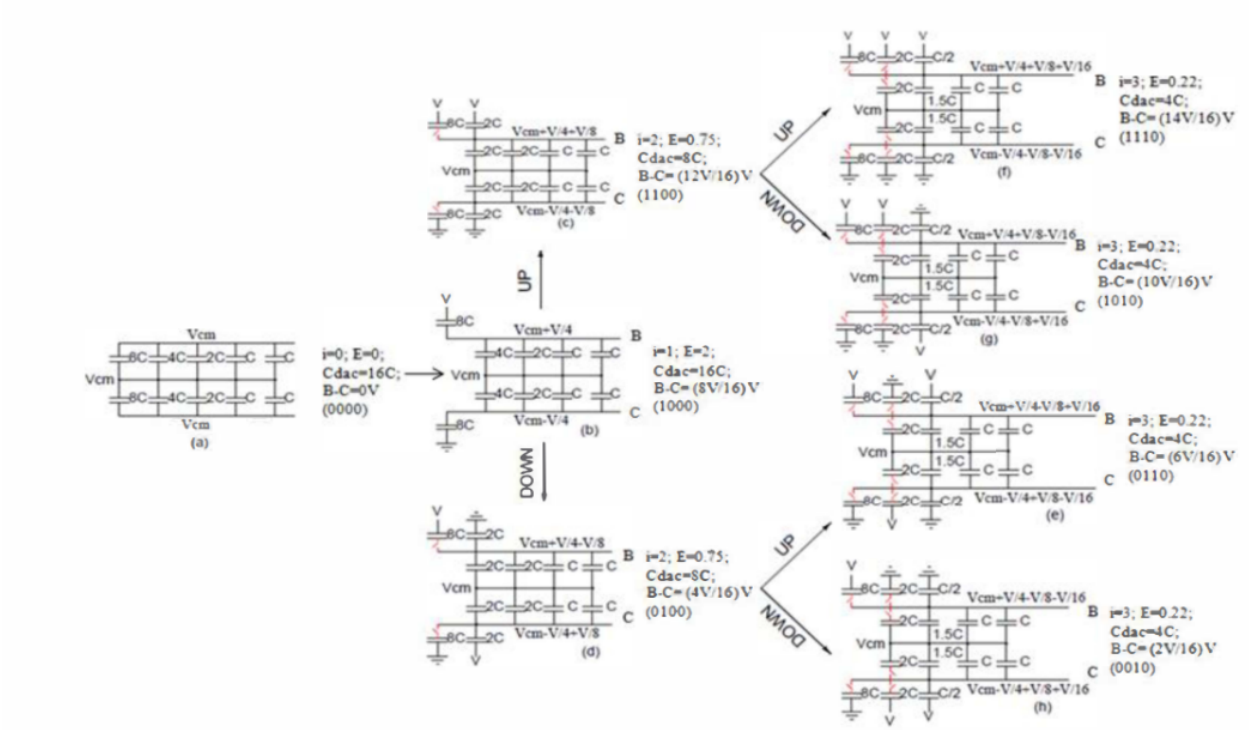


FIGURE 10. Proposed switching technique with quaternary switching capacitors [15].

In conclusion, the literature emphasizes the necessity for energy-efficient solutions in ADC designs, especially for applications like neural prosthetics. QCS introduced in this study significantly enhances existing switching methods by decreasing energy consumption, rendering the DAC energy independent of the output digital code. This technology constitutes a significant advancement in the creation of low-power SAR-ADCs for brain recording front-end interfaces, along with the prevailing trend of energy-efficient solutions in biomedical applications.

This study by Banik et al. [16] establishes the design and implementation of a low-power 6-bit SAR ADC to function at a supply voltage of $1V$, which is standard for low-power biomedical devices. This aligns with contemporary trends in ADC design, wherein researchers prioritize the optimization of power consumption while sustaining requisite performance levels for signal processing. A supply voltage of $1V$ guarantees low power consumption by the ADC, with simulation results revealing a total usage of merely 77.26μW, even at a sampling frequency of $1MHz$.

A fundamental element of the SAR ADC is the S/H circuit shown in Fig. 11, which is

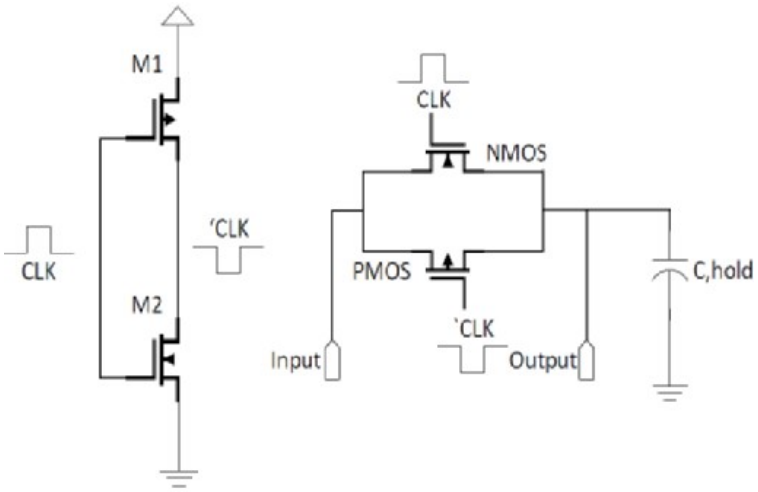


FIGURE 11. Sample and hold circuit [16].

crucial for recording and retaining the input signal during the conversion process. A novel CMOS switch is introduced, enhancing performance and reducing power consumption, as opposed to the conventional transmission gate-based CMOS switch.

The comparator's design shown in Fig. 12 is essential for the SAR ADC's overall performance. Banik et al. [16] describe that the metal-oxide-semiconductor field-effect transistors (MOSFETs) in the comparator function at the sub-threshold region to reduce power consumption. This approach has been extensively employed in low-power systems to attain substantial reductions in current consumption while preserving the necessary functionality.

The SAR logic unit shown in Fig. 13, responsible for executing the binary search logic in the approximation process, is constructed of two shift register blocks, each consisting of asynchronous DFFs. This design guarantees that the conversion process is executed with optimal efficiency, while reducing power consumption.

Furthermore, the binary weighted resistor DAC employed in the design is shown in Fig. 14, comprising resistors of differing values, enhances power efficiency and optimizes settling time relative to alternative DAC architectures, such as the R-2R DAC.

A two-stage operational amplifier (OPAMP) shown in Fig. 15 is employed to augment the

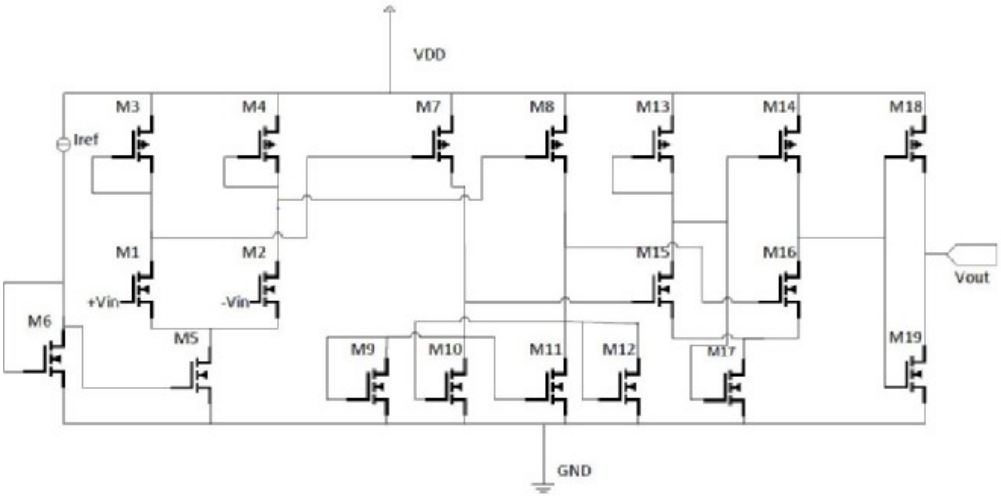


FIGURE 12. CMOS comparator circuit [16].

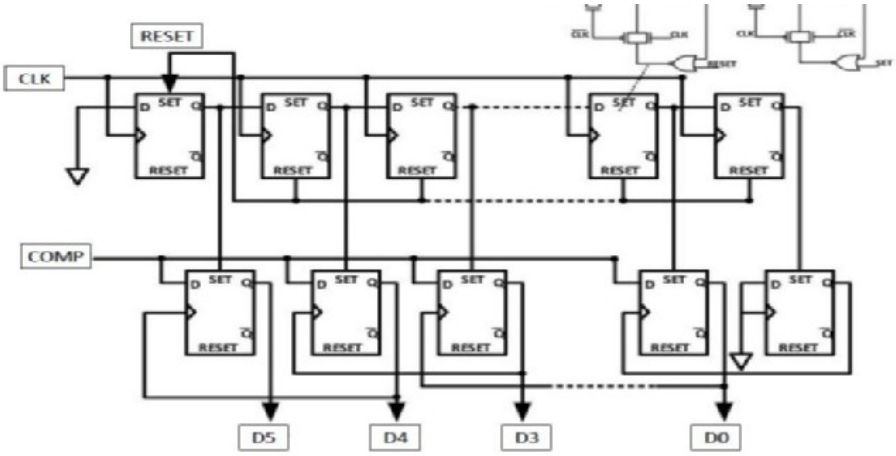


FIGURE 13. SAR unit schematic in cadence [16].

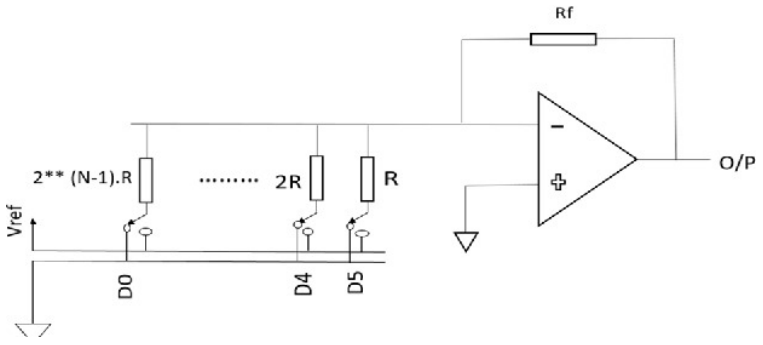


FIGURE 14. Topology of a binary weighted DAC [16].

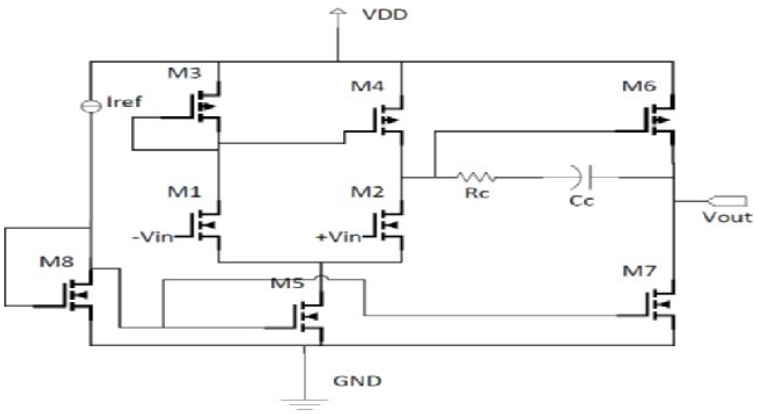


FIGURE 15. Schematic of two-stage OPAMP [16].

DAC's performance, accompanied by a feedback resistor to regulate the output voltage. The OPAMP is meticulously engineered to satisfy essential specifications, such as minimal power consumption, noise margin, phase margin, and common-mode rejection ratio (CMRR). The implementation of a meticulously designed operational amplifier design guarantees that the SAR ADC upholds superior accuracy and stability while utilizing low power.

Simulations for the proposed design were conducted in the Cadence Analog Design Environment (ADE) utilizing 90nm CMOS technology. Banik et al. [16] employed Cadence Spectre and Virtuoso tools to validate the circuit functionality and assess power consumption along with other performance metrics. Their findings indicated that the design satisfies the necessary parameters for biomedical applications, exhibiting an SQNR of 37.34dB and an ENOB of 5.91.

In conclusion, the research conducted by Banik et al. [16] offers a viable approach for low-power SAR ADCs designed for biomedical applications. The implementation of a 1V power supply, an innovative CMOS switch design, and a sub-threshold comparator markedly decreases power usage while preserving satisfactory performance standards.

The design employs a binary weighted resistor DAC and a two-stage OPAMP, hence augmenting its power efficiency. The design has commendable power consumption and performance characteristics; nonetheless, there exists possibility for future enhancements in power efficiency, which could further augment its applicability in biological contexts. This work advances

the research on low-power ADC design for medical devices, emphasizing the importance of the energy economy.

With the rising demand for low-power, high-performance circuits, numerous researchers have concentrated on enhancing R-2R DAC designs to improve resolution, power consumption, and accuracy. Prasad et al. [17] introduces a novel design for a low-power, efficient 4-bit R-2R DAC realized in 45nm CMOS technology, overcoming the constraints of current designs.

The principal issue in R-2R DAC design is the equilibrium of multiple parameters, including power consumption, resolution, accuracy (INL and DNL errors), and speed. Traditionally, R-2R DACs are built using resistors of precise values, and enhancements in performance are frequently attained by altering the operational amplifier utilized in the design.

The article advocates for the implementation of a two-stage operational amplifier, comprising a differential amplifier stage and a gain stage, to improve the performance of the R-2R DAC. This two-stage operational amplifier structure is contrasted with current systems employing a single-stage operational amplifier with less gain, frequently resulting in diminished accuracy and increased power consumption.

Single-stage OPAMP, frequently employed in R-2R DAC, typically exhibit reduced gain, leading to increased errors in INL and DNL. Moreover, the mean power consumption of these designs is typically elevated, rendering them less appropriate for low-power applications.

The proposed architecture in their research considerably minimizes mistakes and power consumption by employing a two-stage OPAMP. The two-stage OPAMP in Fig. 16 comprises a differential amplifier succeeded by a common-source amplifier, facilitating enhanced precision and reduced power consumption relative to single-stage operational amplifier. The suggested design incorporates an R-2R ladder network constructed as illustrated in Fig. 17 with two resistor values, $1k\Omega$ and $2k\Omega$, arranged in series with the binary input pulses (B0, B1, B2, and B3). The design incorporates a feedback resistor of $1k\Omega$ to enhance the performance of the DAC.

The research demonstrates through simulations using the Synopsys H-spice tool that the

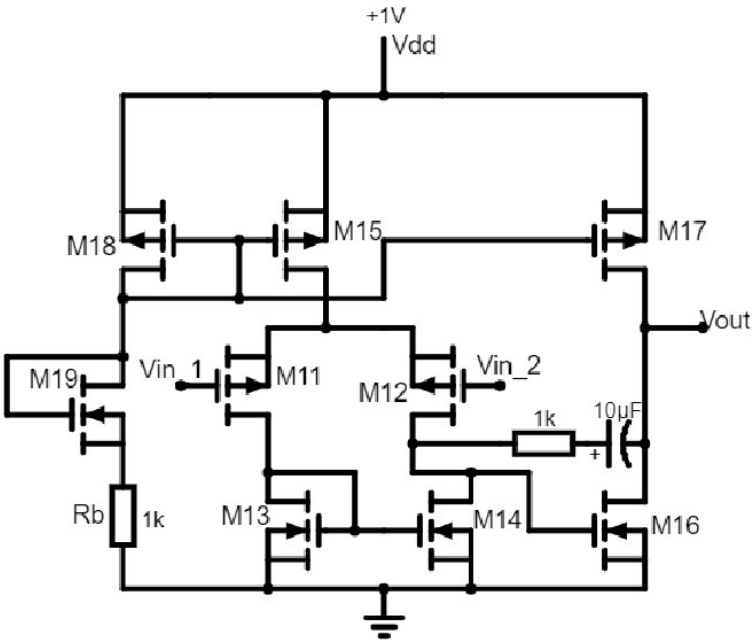


FIGURE 16. Schematic view of op-amp [17].

proposed 4-bit R-2R DAC in 45nm CMOS technology exhibits substantial enhancements compared to current solutions. The average power consumption decreases from 29.4mW to 7.86mW, with static and dynamic powers dropped to below 1mW and 6.429mW, respectively.

Furthermore, the INL and DNL errors are markedly reduced, with values of ±0.13 and ±0.32, respectively, highlighting the efficacy and precision of the suggested design. The suggested two-stage OPAMP-based R-2R DAC presents a viable solution to the issues of power consumption and precision that have historically affected single-stage operational amplifier configurations.

In conclusion, this study significantly contributes to R-2R DAC design by introducing a two-stage OPAMP arrangement that enhances performance regarding power consumption, INL, and DNL errors. The simulation results illustrate the benefits of the suggested design in 45nm CMOS technology, indicating its applicability to low-power applications where accuracy and power efficiency are paramount. The research conducted by Prasad et al. [17] emphasizes the continuous endeavors to enhance DAC performance, crucial for the progression of contemporary electronic systems.

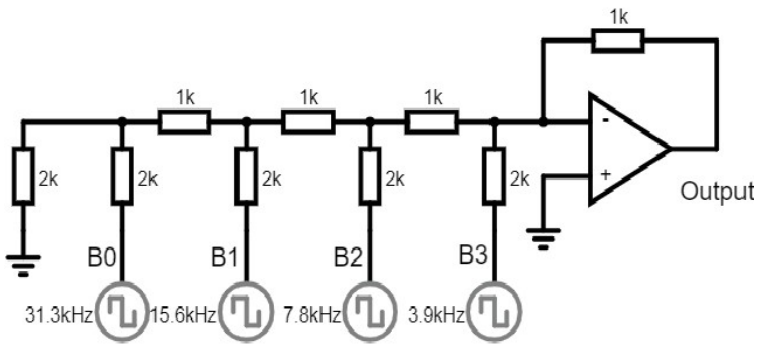


FIGURE 17. Design of 4-bit DAC [17].

The proposed design by Sharma et al. [18] aims to mitigate the leakage power, the primary factor in the total power dissipation of the comparator. To reduce leakage power, the design employs a technique of interpolating two pairs of leakage control transistor sets between the pull-up and pull-down networks in a dynamic single-tail comparator. This aids in minimizing leakage while preserving the comparator's performance. The simulations of the proposed comparator were conducted using Cadence Virtuoso Software using 180nm technology. The simulation findings indicate that the latency of the proposed design was decreased by a factor of 0.87, and overall power consumption is 1.68 times lower compared to the standard dynamic single-tail comparator at an operating frequency of 250MHz.

Sharma et al. [18] also illustrate the configuration of the traditional dynamic single-tail comparator seen in Fig. 18, frequently employed in ADC circuits owing to its zero static power consumption. However, a notable drawback of the circuit is that both outputs utilize identical discharge paths, resulting in an increase in delay. Furthermore, the latch component of the circuit necessitates a substantial current to promptly commence regeneration. Moreover, thermal noise and flicker noise generate random errors in the device. These challenges can be mitigated by employing the right magnitude for accurate detection, but they still remain in traditional designs.

The approach described by Sharma et al. [18] mitigates these constraints by implementing a two-phase comparator with separate paths for the discharge of both V_1 and V_2 outputs. This separation decreases the overall delays of the comparator, providing a substantial enhancement in

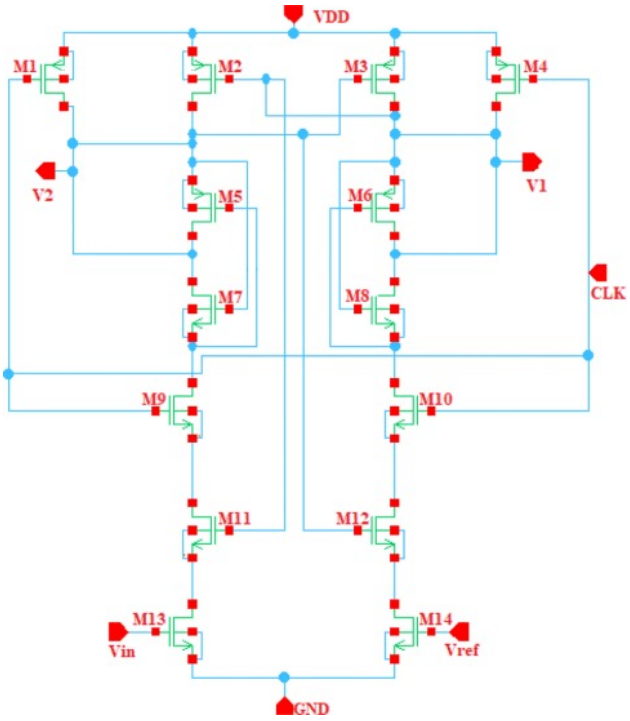


FIGURE 18. Dynamic single tail comparator [18].

performance compared to traditional designs. The proposed design dissipates 40% less power than the typical circuit, achieving an improved value of 80.32μW at a 1.8V supply.

A detailed analysis of the parameter has been conducted concerning power, delay, power-delay product, and the relationship between input voltage differential and power. The findings indicate that power at a 1.8V supply decreases by nearly 40%, with a value of 34.383μW, while delay has been reduced to 8%, with a value of 21.55ps at a 1.0V supply. Voltage scaling simultaneously decreases power consumption and delay, thereby improving the overall performance of the suggested system in terms of power efficiency and speed.

In their study, Mehra et al. [19] present a novel logic system, referred to as "D Logic", shown in Fig. 19, which achieves an 18.53% reduction in the number of transistors needed for SAR logic implementation, alongside a decrease in power dissipation. The 10-bit ADC has been developed utilizing a 90nm CMOS process with Cadence Virtuoso.

The operation of the proposed SAR Logic utilizing D logic is described as follows: at the

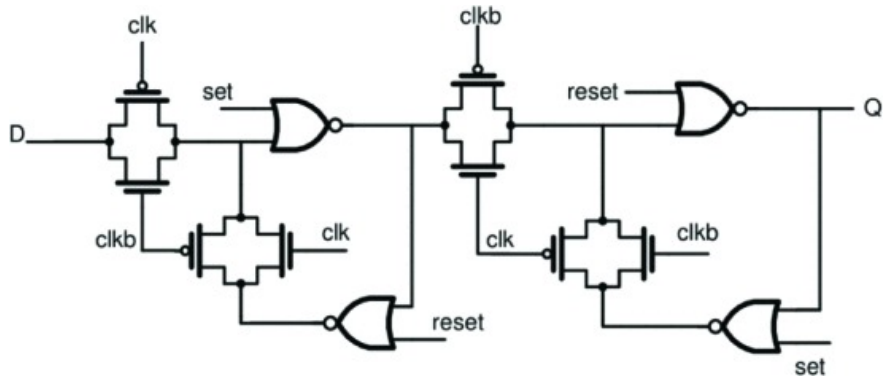


FIGURE 19. Proposed D logic [19].

outset, when the reset signal is active, the MSB flip-flop within the shift register is activated while the remaining flip-flops are deactivated. This results in the output of the MSB flip-flop (Qn) being high, while all other outputs stay low. The Qn is provided to the OR gate of the D Logic utilized in the proposed SAR Logic, resulting in the output Dn being high while all other outputs from Dn-1 to D0 stay low. During the subsequent clock cycle, as the output Qn-1 transitions from low to high, the latch responsible for the first output, specifically the MSB, becomes activated. This activation permits Dn to relay the value of the comparator at the output.

The entire conversion cycle is characterized by the continuation of this process. The proposed SAR ADC designed with "D logic" demonstrates commendable performance, achieving an ENOB of 9.177 and an SNDR of 58.2dB.

Mehra et al.'s [19] study demonstrates a notable reduction in the number of transistors in comparison with traditional SAR ADCs. This document details the implementation of a 10-bit SAR ADC functioning at 20MS/s with a supply voltage of 1V, utilizing a 90nm CMOS process. A power consumption of 67.53μW has been attained. The ADC utilizes a split array DAC, a double tail comparator, and innovative SAR control logic that incorporates an alternative logic system in place of the DFFs typically found in conventional SAR logic. The double-tail comparator implemented minimizes power dissipation in contrast to the traditional dynamic comparator. A DFF is incorporated into the output of the comparator to latch the comparator's output prior to its

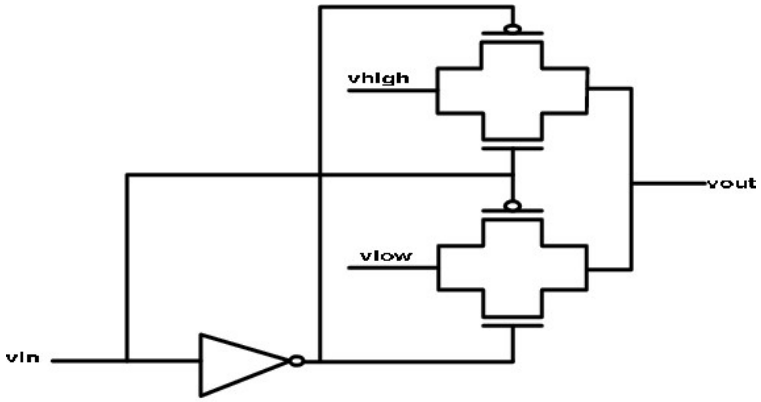


FIGURE 20. 2:1 Multiplexer [20].

transmission to the SAR logic.

In their work, Kulkarni et al. [20] presented a charge-scaling DAC developed in 90nm CMOS technology, constructed with a split array architecture shown in Fig. 21. In order to achieve a compact design, the DAC uses binary-weighted capacitors of $20fF$, $40fF$, and $80fF$. This minimizes the total capacitance area while providing excellent resolution. A 2:1 MUX shown in Fig. 20, which streamlines switching operations and guarantees efficient charge redistribution—a crucial component of SAR ADCs—is a noteworthy design breakthrough.

A two-step split capacitive array DAC for a 14-bit SAR ADC is presented by Savitha et al. [21], as shown in Fig. 22, with a focus on improved resolution and lower power consumption. In order to reduce layout area and power dissipation without sacrificing accuracy, the DAC splits the capacitor array into smaller parts, lowering the overall capacitance and switching energy.

By correcting mismatch faults, capacitive calibration techniques increase the SNDR and ENOB. By increasing the SNDR from 55.3dB to 83.3dB and the ENOB from 8.89 to 13.49 bits, this calibration shows a notable improvement in linearity and performance. It is noteworthy that the system can achieve excellent resolution with only 12μW of power consumption, highlighting its applicability for precise biological applications.

Both approaches place an emphasis on low power consumption and a smaller layout area, which is in line with the needs of biomedical applications that require compact and energy-efficient

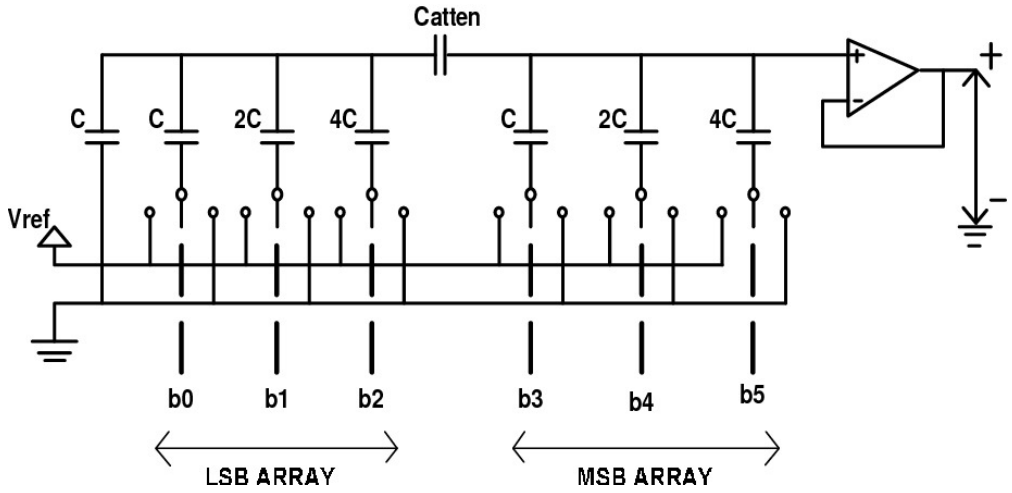


FIGURE 21. Charge-scaling DAC [20].

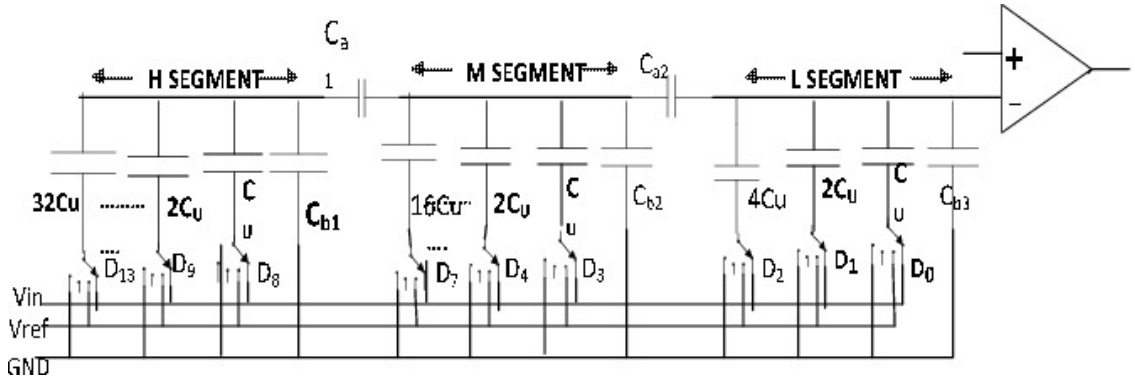


FIGURE 22. Two-split capacitor array DAC [21].

designs. The split array design of the charge scaling DAC provides ease of use and low power consumption, making it appropriate for medium-resolution needs. The two-step split capacitive array DAC, on the other hand, makes things more complicated by using complex calibration methods to boost resolution.

A 2:1 MUX is often used for switching functions, which shows how flexible it is in DAC designs, where it works well to cut down on layout complexity and energy loss. Although efficiency is the top priority for both strategies, their varying resolutions and levels of complexity accommodate a range of application scenarios, from simple monitoring devices to major medical imaging systems.

Finally, the charge-scaling DAC by Kulkarni et al. [20] and the split capacitive array DAC by Savitha and Reddy [21] show how the design of SAR ADCs has improved in ways that work well together. These works provide a solid foundation for making DACs that meet a wide range of biomedical engineering needs, whether the focus is on making them simple and small or on getting high precision and complex calibration. Their contributions highlight the continuous progress in DAC technologies, paving the way for more precise and efficient signal conversion systems.

Presenting a comprehensive SAR ADC design, Hosur et al. [22] emphasize the successive approximation mechanism's working principle. Using this method, the SAR sequentially compares the analog input signal with reference levels to iteratively compute the value of each bit.

Hosur et al. [22] highlight how crucial it is to reduce the power consumption of the digital circuitry in SAR ADCs, which is accomplished by using an optimized design structure. In contrast to the traditional requirement of 2N flip-flops, the authors' non-redundant SAR architecture requires just N flip-flops for an N-bit ADC, reducing the number of flip-flops needed for operation. This reduction demonstrates the usefulness of the suggested methodology by drastically lowering the dynamic power consumption and the overall circuit area.

Hosur et al. [22] also compare the sequence/code register SAR structure and the non-redundant SAR structure shown in Fig. 23 and Fig. 24 in their paper. Their results showed that the non-redundant design achieved a dynamic power consumption of 49.2569μW for a 10-bit SAR ADC implementation, which is around 22.35% lower than the 63.4359μW used by the sequence/code register-based SAR structure.

These energy savings demonstrate how feasible the suggested design is, particularly for applications where power efficiency is essential. The contribution of Hosur et al. [22] is especially relevant for systems that are portable and energy-constrained, such as wearable biomedical monitoring and Internet of Things (IOT) devices.

Hosur et al. [22] guarantee that the produced SAR ADC satisfies the requirements of decreased area, low power dissipation, and sufficient resolution by utilizing the 0.18μm technology.

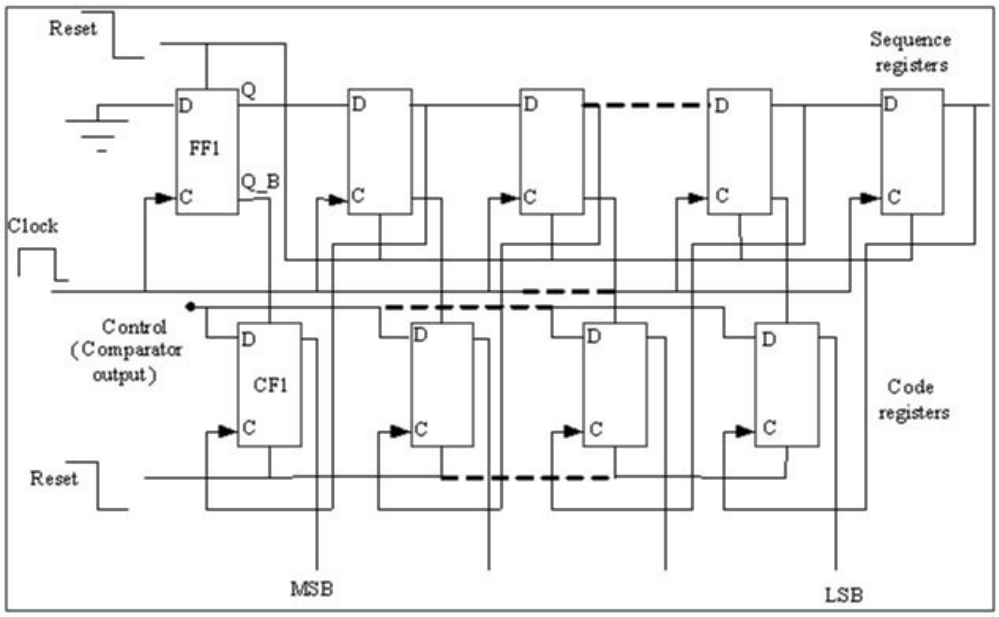


FIGURE 23. Sequencer or code register structure [22].

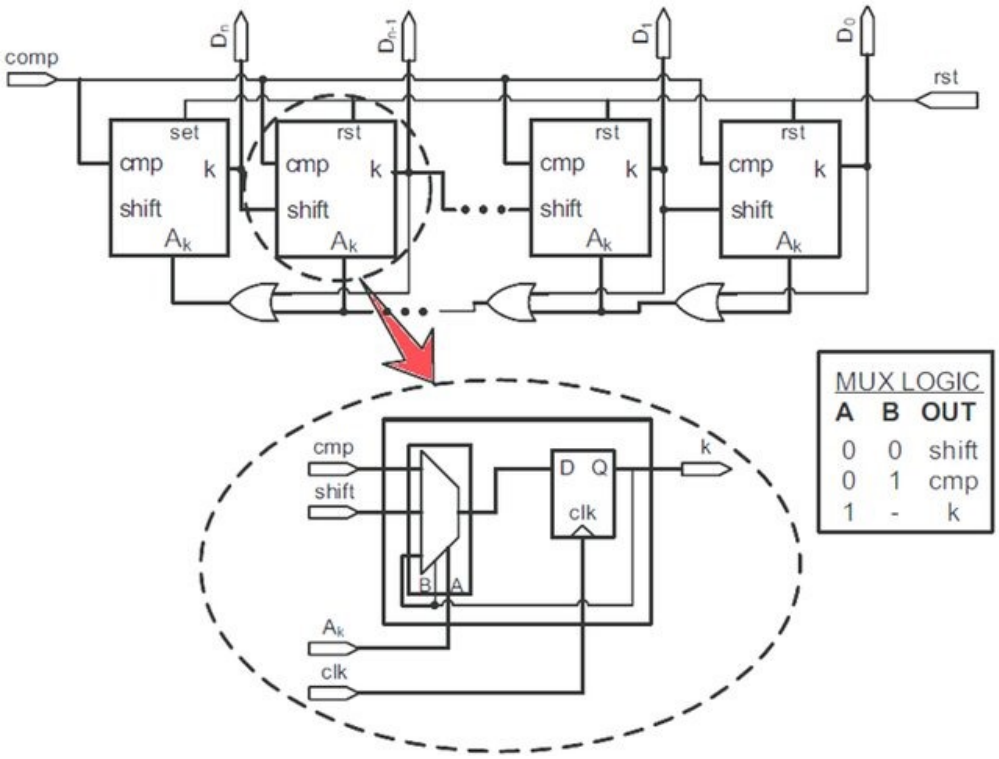


FIGURE 24. Non-redundant SAR structure [22].

This technique supports the viability of deploying CMOS-based SAR ADCs in small systems by offering advantages including lower leakage currents, excellent reliability, and the ability to integrate with other mixed-signal components.

Additionally, Hosur et al. [22] emphasize how scalable their methodology is. Without appreciably raising power consumption or circuit complexity, their architecture might be modified for high-resolution ADCs (more than 10 bits). Due to its scalability, the suggested ADC design is a desirable choice for applications requiring high accuracy, including environmental monitoring devices or medical imaging systems.

In conclusion, Hosur et al. [22] offer a thorough and valuable advancement in the domain of SAR ADCs. It demonstrates how non-redundant structural enhancements might propel substantial applications in industry, healthcare, and the environment by enhancing energy efficiency and spatial optimization. This publication's technique, supported by research, provides a vital reference for academics and engineers engaged in low-power ADC design.

Table 2 shows the comparison of this thesis with previous works on SAR ADC.

TABLE 2. Comparison of Existing SAR ADCs

References	[1]	[14]	[19]	[23]	[24]	[25]	[26]
Technology	90nm	90nm	45nm	90nm	90nm	180nm	180nm
No. of Bits	8	8	10	8	10	10	8
Speed	100KS/s	10KS/s	20MS/s	250MS/s	1MS/s	200KS/s	100KS/s
Power Supply (V)	1.2	1.0	1.0	1.2	1.2	1.0	1.8
Power	20μW	400nW	67.53μW	0.98mW	11.88μW	4.45μW	0.05mW

CHAPTER 3

DESIGN METHODOLOGY

The SAR architecture, illustrated in Fig. 1, has four essential components: S/H circuit, DAC, SAR Logic and the comparator. The analog input initially enters the S/H circuit, which retains the input signal during the conversion process. The sampled signal is promptly transmitted to the comparator. The comparator thoroughly evaluates the signal against a certain reference value, which is the precise analog output from the DAC. The comparator generates a digital output, either logic high or logic low, depending on whether the sampled input exceeds or falls short of the DAC outputs.

This work provides details on the architecture of a 10-bit ADC developed in a 45*nm* CMOS process utilizing Cadence Virtuoso and the GPDK045 technology library.

3.1 Sample and Hold

The S/H circuit is an essential component of any SAR ADC architecture, it functions as the preliminary stage when the analog signal is introduced into each configuration. The primary responsibility is sampling incoming voltage and maintaining it at a consistent level during conversion. This circuit is essential for accurate ADC conversion, as it mitigates any fluctuations in the analog signal over time by providing a stable voltage for evaluation.

3.1.1 Pass-Transistor Switch

The schematic for the pass-transistor switch utilized in this project has been carefully designed to reduce the significant impact of charge injection ([16] and [19]). To accomplish this goal, both specific PMOS and NMOS devices are incorporated in a certain way within the structure. The charge introduced into the sampling capacitor, C_H, via those transistors is entirely eliminated in an effective manner. This cancellation works for a single input level, ensuring reduced errors in the sampled signal. However, while this design mitigates charge injection, it does not completely eliminate the effects of clock f eedthrough. Clock feedthrough occurs when the clock signal transitions through the MOS switch, coupling to the sampling capacitor through the gate-drain or gate-source overlap capacitance. This overlap introduces an error because the overlap capacitances

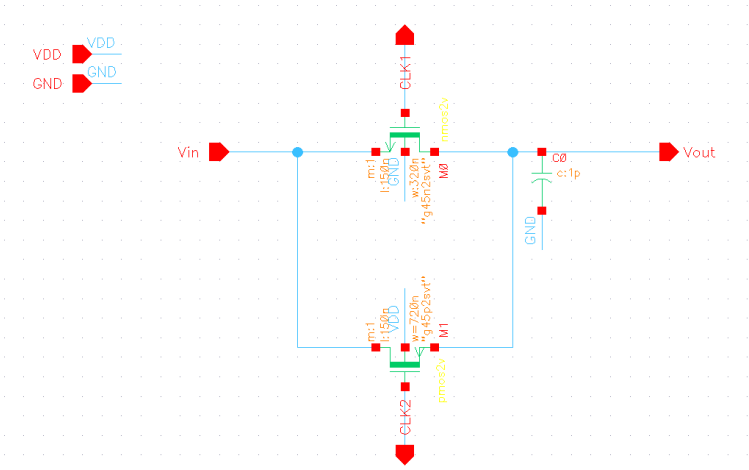


FIGURE 25. Pass-transistor switch.

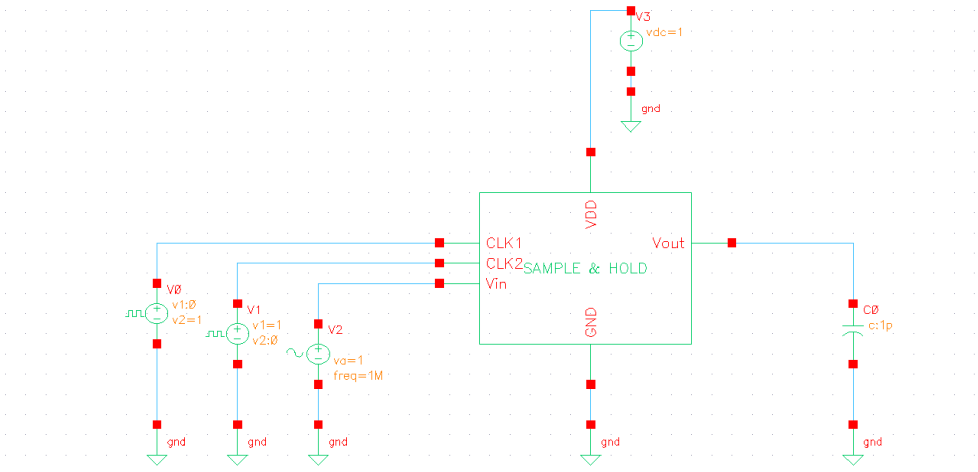


FIGURE 26. Testbench for sample and hold circuit.

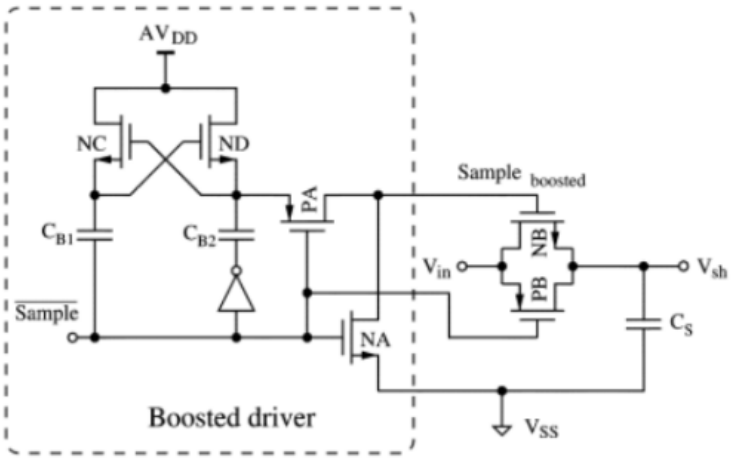


FIGURE 27. Schematic of track and hold with boosted driver [27].

of the NMOS and PMOS transistors are not identical, resulting in incomplete cancellation of the injected charge.

A significant drawback of transmission gate-based S/H circuit designs is the subthreshold leakage current that occurs when the switch is OFF, leading to fluctuations in the output voltage. This leakage is primarily caused by the ON resistance between the source and drain terminals, which is influenced by the input clock signal. To mitigate this issue, the clock signal can be boosted to reduce the ON resistance. The ON resistance of an NMOS sampling switch is given by:

$$R_{ON} = \frac{1}{\mu_n C_{OX}(V_{DD} - V_{in} - V_{TH})\frac{W}{L}}$$

R_{ON} is dependent upon the input voltage, V_{in}. Any variation in input voltage affects R_{ON}, resulting in nonlinearity [28]. In this instance, sampling is poorly performed due to charge leakage when the switch is in the OFF state. Consequently, it negatively impacts the linearity of the signal.

The circuit responsible for boosting the clock signal is known as a boosted driver. The schematic of the S/H circuit with a boosted driver, followed by the transmission gate, is shown in Fig. 27 [27].

In this design, the driver circuit is connected to the gate of the NMOS transistor in the

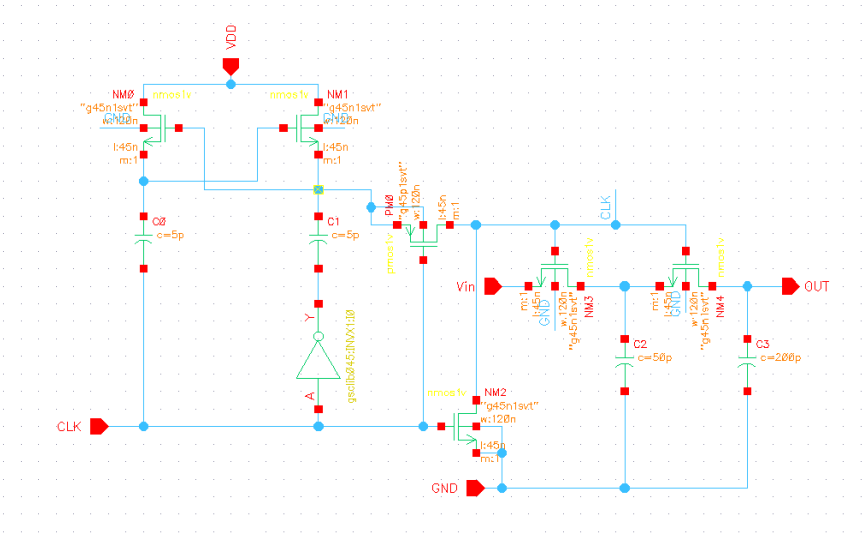


FIGURE 28. S/H circuit with cascaded simple switches and boosted driver.

transmission gate. The boosted voltage, which ranges between $2V_{DD} - \Delta V$ and ground, is observed at the output due to charge sharing between the PA and the parasitic capacitance at the output node [27],[29]. The boosted voltage can be calculated as follows:

$$V_{\text{boosted}} = 2V_{DD}\left(\frac{C_{B2}}{C_{\text{gateNMOS}} + C_{B2} + C_{\text{parasitic}}}\right)$$

3.1.2 Cascading Simple Switch with Boosted Driver

In ADC front-end circuits, achieving uniform conductance is essential. To meet this requirement, bootstrap circuits are employed. These circuits generate a linearly varying output voltage based on a variable input signal. Typically, clock signals drive these bootstrap circuits. In this design, the sample signal applied to the capacitors C_1 and C_2 (the boosted capacitors) ranges from 0 to V_{DD}. An inverter is placed before the capacitor C_2. By cascading two simple NMOS transistors, the boosted driver achieves low power consumption and wide bandwidth.

Initially, both capacitors C_1 and C_2 are uncharged. When the sample bar signal is low, the inverter's output voltage is approximately V_{DD}. At this point, the top plate of C_1 is at $0V$, while the top plate of C_2 is at V_{DD}. Consider transistor T_1: with the gate and drain at V_{DD}, the source

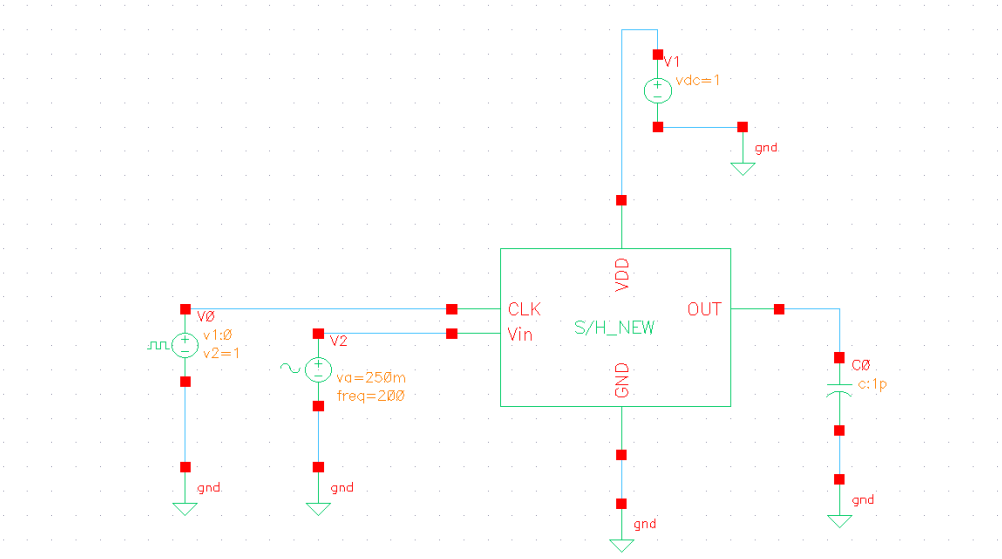


FIGURE 29. Testbench for cascading simple switch with boosted driver sample and hold circuit.

potential is $V_{DD} - V_{TH}$. When the sample bar signal transitions to high (V_{DD}), the inverter output and the top plate of capacitor C_1 both shift to V_{DD}, while the voltage at C_1's top plate becomes $2V_{DD} - V_{TH}$. This voltage, $2V_{DD} - V_{TH}$, is applied to the gate of transistor T_2. With the drain at V_{DD}, the source voltage of T_2 rises towards V_{DD}, causing the capacitor to charge towards V_{DD}. In the next phase, the gate of transistor T_1 is set to $2V_{DD}$, with the drain still at V_{DD} and the source potential at V_{DD}. After a few cycles, both capacitors C_1 and C_2 are fully charged to $2V_{DD}$.

The schematic for the S/H circuit with cascaded simple switch design and boosted driver is shown in Fig. 28. In this design, cascaded transistors T_5 and T_6 act as the sample switch. The boosted voltage from T_3 controls the cascaded switch, removing the restrictions imposed by low supply voltage and enabling rail-to-rail input signal handling through the two NMOS switches [27]. This circuit consumes less power than the pass transmission gate shown in Fig. 25. The transistors are sized as $12\mu m/0.36\mu m$, and the capacitors have the following values: $C_{S1} = 200pF$, $C_{S2} = 50pF$, and $C_1 = C_2 = 5pF$. Since the capacitor values for C_{S1} and C_{S2} differ, the charging and discharging time constants also vary, ensuring a stable output. This results in constant sampling,

improved linearity, and reduced leakage.

The suggested booster clock S/H circuit designs demonstrates improved power efficiency and enhanced noise immunity for the sampling of biological signals but have a drawback in that their sampling frequency is only about $250kHz$(low frequency), which is much slower than the SAR ADC's target sampling rate of SAR ADC. The S/H circuit must function at a sampling rate within the intended range in order to guarantee compatibility with the entire SAR ADC block. In order to solve this problem, we suggest a different circuit design.

3.1.3 Proposed Sample and Hold Circuit

A dummy switch that is managed by an inverted clock is used to reduce charge injection errors [24] and [30]. As seen in Fig. 30, this switch is strategically placed to absorb any remaining charge from the sampling switch. By successfully minimizing undesired glitches during operation, the system enhances signal integrity and guarantees a more seamless conversion process. The inverted clock is also produced by an inverter, which makes it possible for this new circuit to operate correctly. This circuit arrangement reduces the overall on-resistance of the transmission gate, ensuring better performance during switching.

In my thesis, I initially used a pass transistor switch for the S/H circuit. Subsequently, I have incorporated the S/H circuit from [27],[29]. The final design makes use of a S/H circuit that functions within the SAR ADC's necessary sampling frequency range. The testbench shown in Fig. 26, Fig. 29 and Fig. 31 tests the basic functionality of the S/H circuit, ensuring that the signal is correctly sampled and held over the required period. The input and control signals (S/H) are applied, and the voltage across the capacitor is monitored during the test.

3.2 Comparator

The comparator is a crucial component of the SAR design since it is responsible for assessing the voltage that the S/H circuit maintain after stabilizing the analog input signal. The comparator produces a binary output by comparing the voltage produced by the inbuilt DAC with a reference value. Whether the sampled input voltage is higher or lower than the reference voltage is

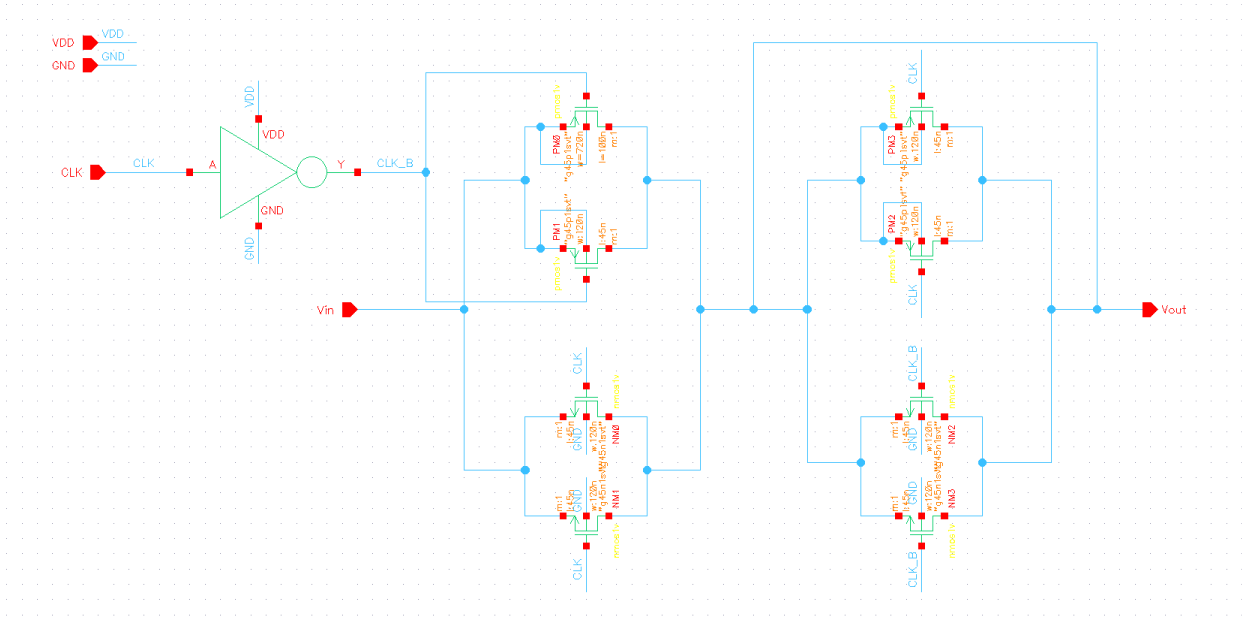


FIGURE 30. Proposed sample and hold circuit.

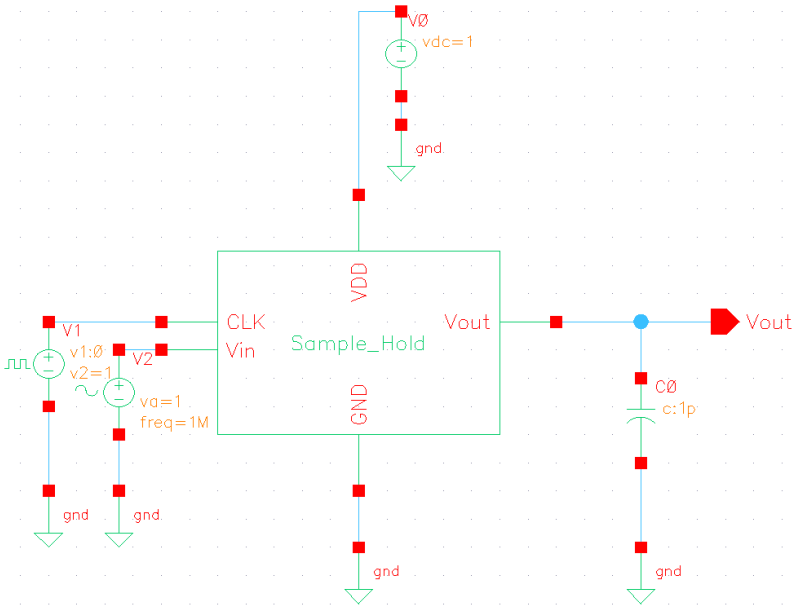


FIGURE 31. Testbench for proposed sample and hold.

shown by this binary output. This comparison's outcome is crucial for directing the SAR operations and figuring out how to iteratively improve the ADC's input voltage approximation.

As an ADC that produces a single bit of digital data, the comparator is a basic component of all ADCs. This bit is then processed by the remainder of the circuit to generate the necessary N-bit data. ADCs can use a variety of comparators, including OPAMP-based, latching, dynamic, and double-tail comparators.

A "dual tail comparator" design referred from [19] and [31] is created when two comparator units are incorporated into a single IC. Based on the respective magnitudes of the two input voltages, this configuration, as shown in Fig. 32, compares them and outputs a digital signal. Testbench for Double tail comparator is shown in Fig. 33.

The current source or current mirror that is used to bias the circuit is often indicated by the term "tail" in the name. To provide fast latching independent of the common mode voltage, a dual tail comparator circuit was used, which permits both a large current in the latching stage and a small current in the input stage. The input stage still requires a significant common-mode voltage, which may be problematic at low supply voltage. Additionally, a differential pair of clock signals is required in the double-tail dynamic comparator, adding power and space requirements.

The operation of the double-tail comparator is characterized by several key phases. During the reset phase, when CLK is low and M11 and M0 are inactive, transistors M3 and M12 precharge the t_n and t_p nodes to V_{DD}. This action ensures that transistors M5 and M8 discharge the output nodes OUTN and OUTP to the ground. As the clock signal transitions to V_{DD}, M3 and, M12 turn off, while M11 and M0 activate. At this point, the voltages at nodes t_n and t_p begin to decline at a rate given by, determined by I_{M11}/C$t_{n(p)}$. The intermediate stage of the comparator, involving M5 and M8, contributes to noise reduction and effectively transfers V$t_{n(p)}$ to the cross-coupled inverters, benefiting from strong input-output shielding.

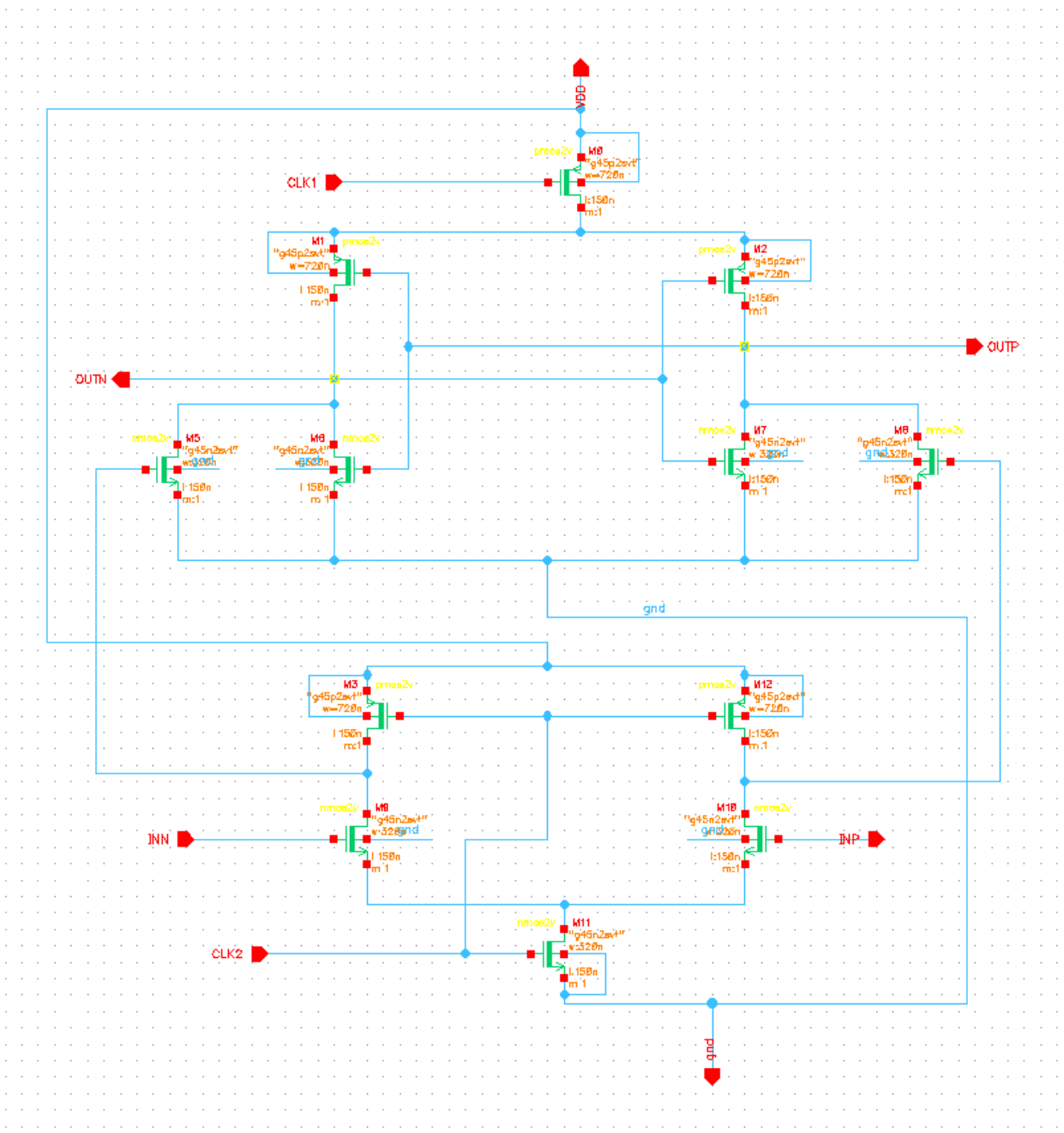


FIGURE 32. Double tail comparator.

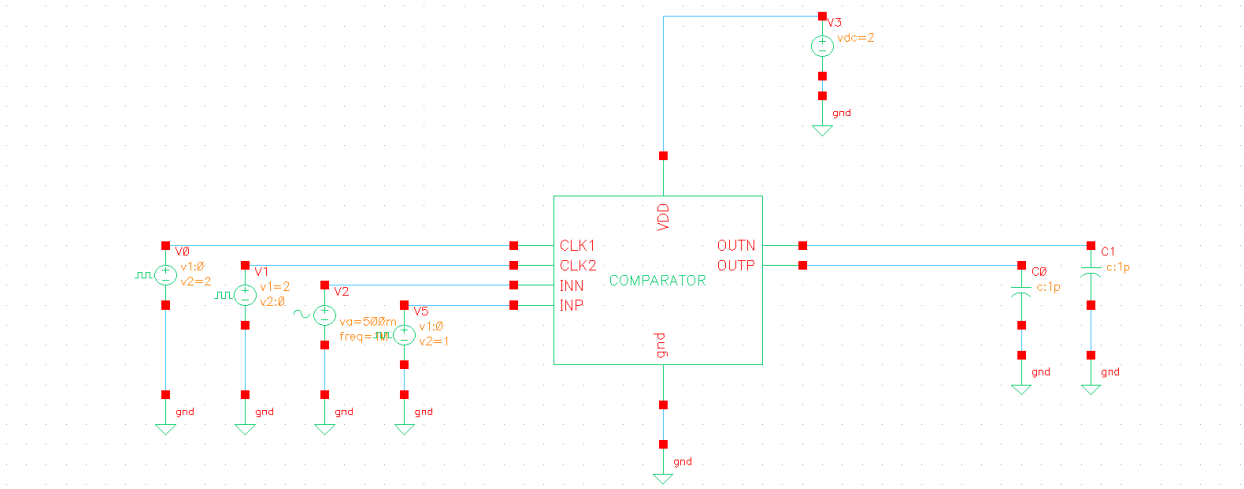


FIGURE 33. Testbench for double-tail comparator.

3.3 Digital-to-Analog Converter

The DAC employed within the SAR ADC assumes a distinctive role. In contrast to stand-alone DACs, which primarily transform digital signals like audio into analog form, the DAC integrated into a SAR ADC undertakes an internal function which is to generate reference voltages. These reference voltages are subsequently pitted against the sampled input voltage for comparison. Driven by the prevailing binary pattern stored in the SAR, the DAC's operation contributes to the iterative process of approximation. As the SAR progressively hones its estimation, the DAC adapts to produce increasingly precise reference voltages.

3.3.1 Charge Redistribution DAC

The charge-scaling DAC is a widely used DAC architecture in CMOS technology [19]. A collateral array of double-weighted capacitors totaling 2^NC is depicted in the Fig. 34. The output voltage, V_{out}, is determined by the voltage division between the capacitors after each one has been first discharged and then charged to either V_{ref} or ground by the digital signal. Since the array of capacitors adds up to 2^NC, a voltage divider between the MSB capacitor and the remainder of the array will occur if the MSB is high and the other bits are low. V_{out}, the analog output voltage, is expressed as follows in the equation:

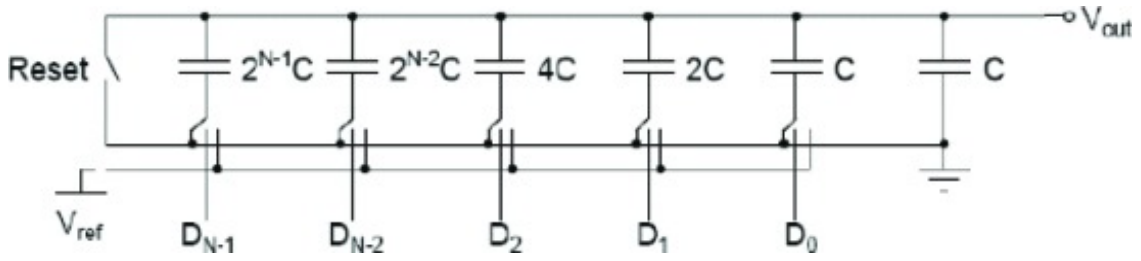


FIGURE 34. Charge redistribution DAC [19].

$$V_{out} = V_{ref}\frac{2^{N-1}C}{2^{N-1} + 2^{N-2} + ... + 4 + 2 + 1 + 1)C} = V_{ref}.\frac{2^{N-1}C}{2^{N}C} = \frac{V_{ref}}{2}$$

which indicates the fact that the MSB modifies a DAC's output to $\frac{v_{ref}}{2}$. For every capacitor, the ratio of V_{out} to V_{ref} can be generalized to

$$V_{out} = V_{ref}.\frac{2^{K}C}{2^{N}C} = 2^{K-N}.V_{ref}$$

where all other bits are believed to be zero and the kth bit, D_k, to be one. The value of V_{out} for any digital input word can then be found using superposition by

$$V_{out} = \sum_{k=0}^{N-1} D_K 2^{k-N}.V_{ref}$$

It is possible to attain a high resolution (10- to 12-bit) without requiring a linear capacitor. The charge redistribution architecture is widely used due to its ease of usage and relative accuracy.

The redistribution of charges Binary-weighted capacitor array arrays need a medium amount of space and have a low power consumption and straightforward design. However, it also has several serious problems. However, as the resolution increases, though, the MSB capacitor's size becomes crucial. For instance, the MSB capacitor needs to be $C_{MSB} = 2^{N-1}.0.5pF = 256pF$ if the unit capacitor is $0.5pF$ and a 10-bit DAC is being created. One method for reducing capacitor size is to use a split array.

3.3.2 Split Array DAC

This research uses a "split capacitor based BWC array" to address the power consumption and area-related concerns that charge redistribution DAC has [11], [20]. In Fig. 35, the suggested DAC structure is displayed. The BWC array is split into two smaller BWC in a split capacitor-based DAC, and a coupling capacitor is positioned in between the two sections.

This circuit scales a voltage reference that the capacitive network provides to transform a 10-bit digital input word into the corresponding analog signal. A capacitive network, switches, and operational amplifier are some of the building parts of Fig. 35, to which the digital word is assigned.

The DAC switch circuitry seen in the Fig. 36 receives the input digital word first. The switch determines the specific voltage to which the capacitor is to be charged based on the logic value of each bit of the word. The switch selects the input that is linked to the "GND" and charges the capacitor to $0V$ if the input bit in the digital word is logic "0". The capacitor is charged to V_{ref}, or $1.1V$, if the input bit in the digital word is logic '1'. The array is divided into an LSB array and an MSB array using an extra attenuation capacitor after the output is removed from a separate node. The leftmost switch is now represented by LSB, D0, and the rightmost switch by MSB, D9. The attenuation capacitor's value can be determined by

$$C_{attenuation} = \frac{\sum \text{LSB Array Capacitors}}{\sum \text{MSB Array Capacitors}}$$

The attenuation capacitor acts as a voltage divider, balancing the charge between the MSB and LSB arrays. This ensures that the LSB's contribution to V_{out} is properly scaled down relative to the MSB capacitors.

The 'terminating capacitor' is the capacitor at the end of the network. The effective resultant analog voltage for each digital combination is determined by the capacitors, which are charged to varying voltages depending on the input digital word.

It is crucial to choose the unit capacitance value. To reduce power consumption, the

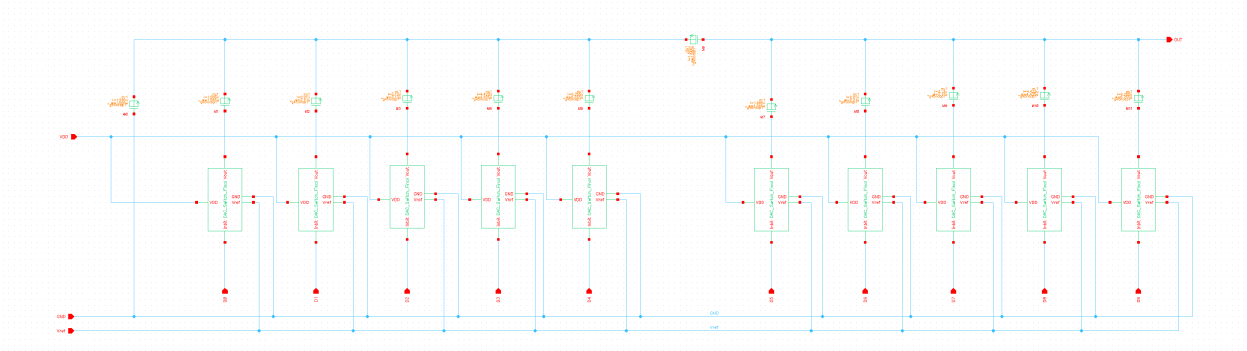


FIGURE 35. Charge-scaling DAC using a split array.

capacitor's value should be as little as feasible. However, the minimal value of the unit capacitance is constrained by design guidelines, thermal noise, and capacitor matching. A MOS capacitors are easily integrated into standard CMOS fabrication, were employed in this experiment, and the lowest value was $36.9fF$. The capacitance values of the capacitor array are displayed in Table 3. Inverters and transmission gates are used in digital logic to make switches.

TABLE 3. Capacitor Values

Device	Size (fF)
C	36.9343
2C	73.6522
4C	147.737
8C	295.276
16C	590.005

3.3.2.1 DAC switch. Using CMOS technology, a DAC switch is very easy to create because it only requires one inverter and two transmission gates. One or both of the transmission gate switches can be enabled using the selection signal (V_{in}). V_{ref} is connected to the output V_{out} when V_{in} is HIGH. GND is connected to the output V_{out} when V_{in} is LOW. The 2:1 multiplexer's from [20] and [32] schematic is displayed in Fig. 36.

3.3.3 Topology of a Binary Weighted DAC

Achieving high conversion accuracy, fast speed, and high resolution is the primary difficulty for all DACs. Because of their straightforward construction, quick settling time, and low power

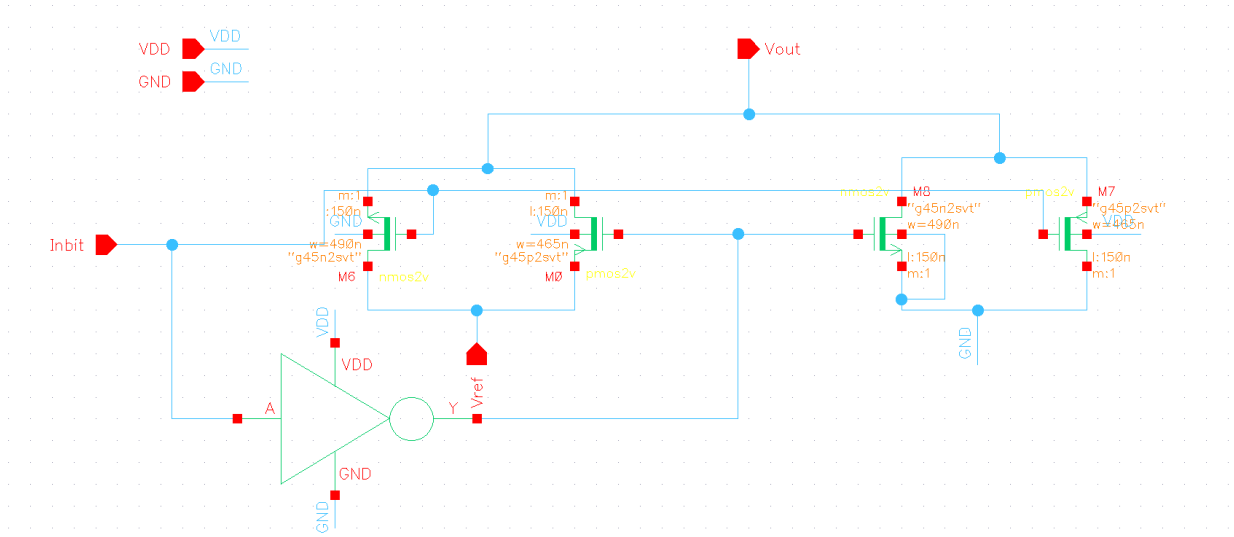


FIGURE 36. 2:1 multiplexer.

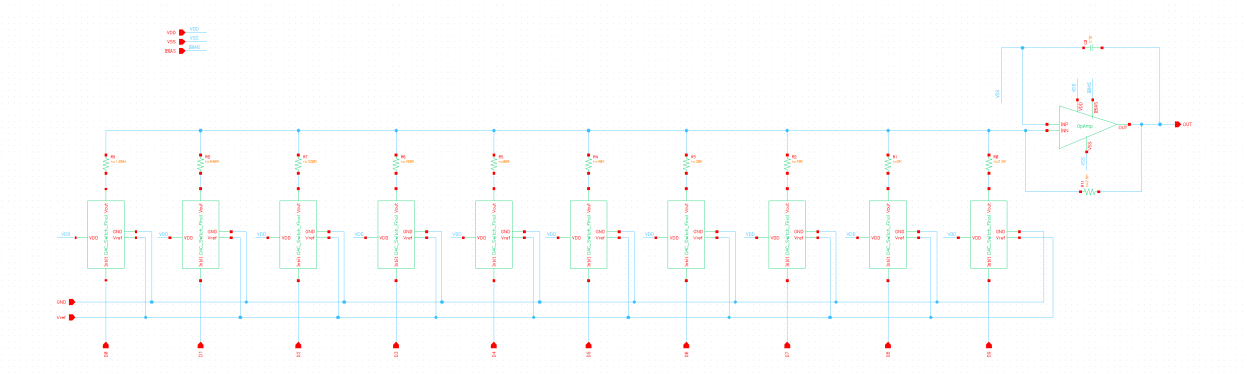


FIGURE 37. Topology of a binary weighted DAC.

consumption, resistor-based DAC architectures such as binary weighted DAC, R-2R DAC, string DAC, etc. are frequently used. However, the precision of the resistors employed and the stability of the voltage reference have a significant impact on the resistor-based DAC's conversion accuracy. In contrast, the switching delay time mostly determines the conversion speed.

Binary weighted resistor architecture shown in Fig. 37 was employed in this study, utilizing 2.5K (MSB), 5K, 10K, 20K, 40K, 80K, 160K, 320K, 640K, 1.28M (LSB), and 2.5K feedback resistors [16].

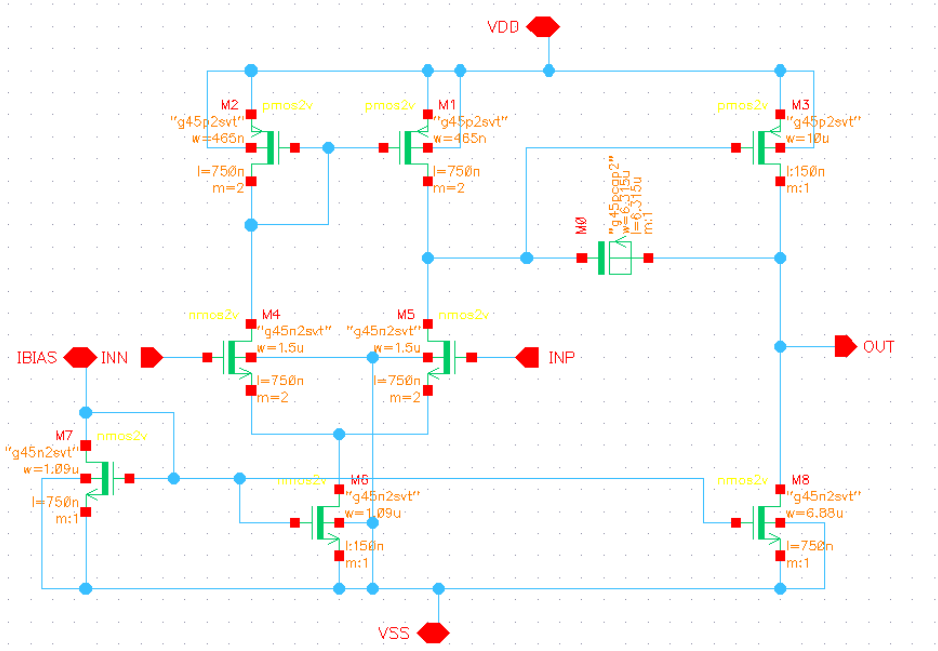


FIGURE 38. Operational amplifier.

3.3.3.1 Two-stage OPAMP. This circuit is made up of two stages, as shown in the schematic(Fig. 38): a conventional differential amplifier for the first stage and a typical common-source amplifier for the second [16] and [32]. The second stage offers a good output swing because there are only two transistors. By creating a structure of miller compensation between the first and second stage outputs, the capacitor M0 improves the stability of this conventional two-stage differential operational amplifier. To supply bias current for first- and second-stage amplifiers, MOSFET M6, M7, and M8 construct a current mirror. Each step of gain can be multiplied to obtain the total gain due to the two-stage structure.

To attain its intended performance parameters—noise margin, phase margin (PM), low power, output gain, unity gain response, CMRR, etc. the OPAMP has been meticulously constructed. The DAC and OPMAP circuit architectures are displayed.

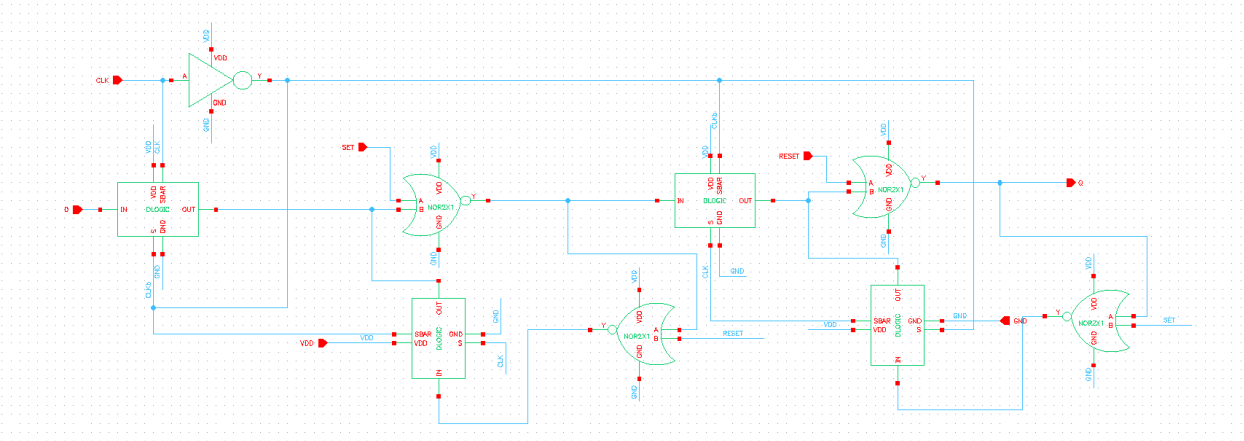


FIGURE 39. D flip flop.

3.4 Successive Approximation Register Logic

The crucial SAR is located at the heart of the SAR ADC. This part, which embodies the fundamentals of the successive approximation process, is what makes the SAR ADC function. Starting from an initial approximation, the SAR refines itself through a series of subsequent comparisons. This corresponds to 10 separate comparisons in the case of a 10-bit ADC, starting at the MSB and ending at the LSB. The SAR changes the bit that is being examined after every comparison. The bit under examination is set to 1 if the DAC's output registers a value less than the input voltage, and to 0 otherwise.

The binary search algorithm is implemented using SAR ADC employing SAR control logic. SAR control logic uses the comparator's result to successively determine the bits' values. SAR is in reset mode and all outputs are 0 for the first clock cycle. Data conversion and sequential bit determination take place during the course of the following 10 clock cycles. The results of the full conversion are stored in the final cycle.

A shift register and a code register are the first and second rows of positive edge-triggered DFFs with set-reset, respectively, used in conventional SAR logic. The first row of DFFs receives a common clock signal, and the output of the subsequent DFF is used as the input clock signal for the current DFF in the second row. The input to the subsequent flip-flop in the shift register is the

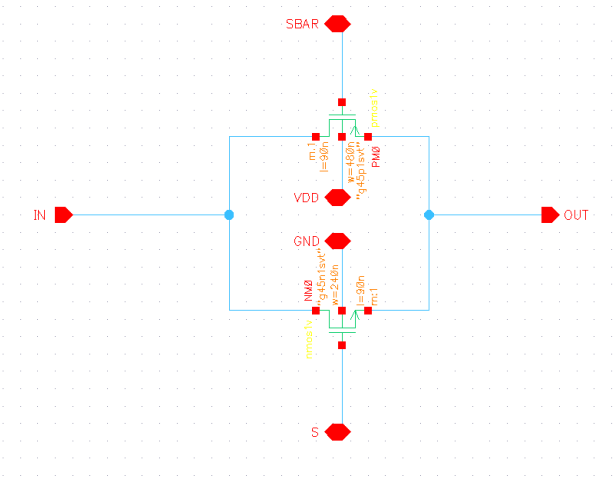


FIGURE 40. D logic switch.

output of each individual DFF. Conversely, the code register receives its input from the comparator's output. Additionally, the output of the flip-flops in the code register provides the final output of the conventional SAR logic, while the output of the flip-flops in the shift register acts as the set signal for the flip-flops in the code register [11] and [16].

One of the sequencer's outputs sets a flip flop in the code register for every clock cycle. The sequencer sets the flip flop's output, which serves as a clock signal for the prior flip-flop. The results of the full conversion are stored in the final cycle.

Synchronous with the internal clock, the shift register at the top serves as a sequencer. The conversion value is kept in the bottom register. Since the circuit functions flawlessly and is quite simple to implement, we have designed a DFF based on transmission gates and NOR gates.

The output's MSB bit will be initially set to "1" and the other bits to zero when the circuit is operating. The DAC then transforms the digital output into an analog value that is comparable to the SAR output. The DAC output voltage (V_{out}) is compared to the original sampled analog value (V_{in}). We must adhere to two rules in order to fully understand its operating concept. To begin with, the MSB bit is originally set to "1". The initial output of a 10-bit ADC, for instance, is 1000000000. Second, the first bit, which was set to "1", will change to "0" and the second bit will be "1"

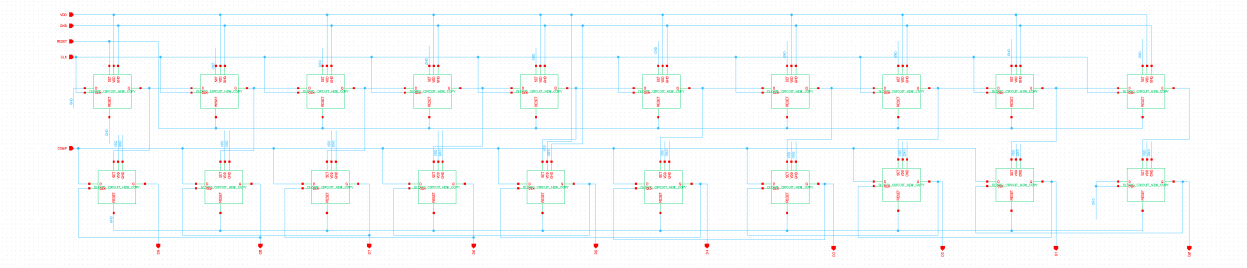


FIGURE 41. Conventional SAR logic.

(0100000000) if $V_{in} < V_{out}$. However, the following bit will be set to "1" (1100000000) if $V_{in} > V_{out}$, and this procedure keeps on until the data conversion is complete. We have created an analog comparator for the purpose of comparing two values. For an N-bit ADC to fully convert one sample's data, N clock cycles will be needed.

Reducing the number of transistors utilized in the SAR logic helps reduce the area occupied. This work proposes a novel logic that uses less transistors and substitutes the DFFs in SAR Logic. Fig. 39 displays the circuit diagram for the suggested logic, which will be referred to as "D Logic."

This DFF has active low SET & RESET and operates asynchronously. We have developed a DFF utilizing transmission gates(shown in Fig. 40) and NOR gates shown in Fig. 39, as it is simple to put in and the circuit functions precisely [11]. Flip-flops based on transmission gates are utilized for low-power applications.

Transistors of minimum size and double length are selected to enhance the power efficiency. Low-V_T transistors are employed in the critical path and high threshold voltage transistors in the non-critical paths to further reduce the leakage power while preserving speed. Thus, high performance Flip-Flops are provided by this dual threshold method.

Overall, the SAR design has the benefit of being able to be expanded to greater resolutions simply by extending the shift registers because the logic is iterative. Additionally, there are fewer signal changes overall, which reduces energy usage. The supply voltage can be decreased to further reduce power consumption.

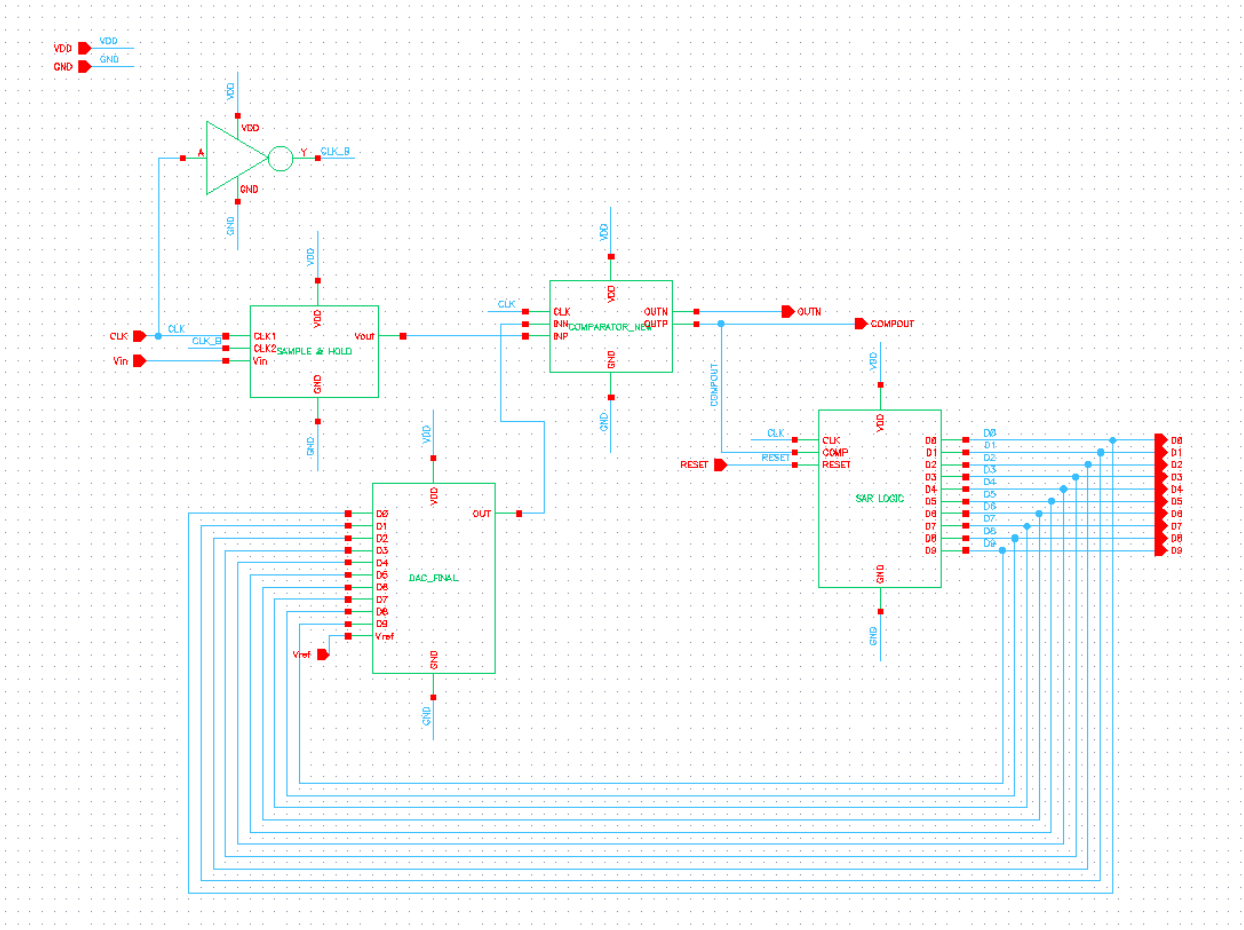


FIGURE 42. Successive approximation register ADC.

3.5 Successive Approximation Register Analog-to-Digital Converter Block

The SAR ADC uses a 45*nm* production node and is precisely designed for biological applications. A S/H circuit, comparator, DAC, DFF, and SAR control circuitry are all part of its simple yet reliable architecture.

With the use of successive approximation methods, the SAR ADC effectively transforms analog inputs into digital codes, making it the best option for low-power applications. It computes every bit of the output using a binary search algorithm in each cycle, improving the approximation with each iteration. Since this repetitive process minimizes power consumption while enabling exact conversion, SAR ADCs are particularly well-suited for use instances where power is a concern.

The integrity of the ADC process and efficient component cooperation are ensured by the smooth integration of each block into the SAR ADC design(shown in Fig. 42) to carry out its assigned duty. The SAR ADC goes through extensive functional testing after integration to confirm each block's functionality. To ensure that the ADC output can precisely transform analog signals into digital pulses while maintaining the frequency and amplitude of the original signal, it is carefully simulated.

CHAPTER 4

SIMULATION RESULTS AND ANALYSIS

In 45nm CMOS technology, the suggested SAR ADC has been developed using Cadence Virtuoso. For each sub-block, a transient analysis has been conducted independently. Biomedical signals, such the ECG signal, range in amplitude from 1mV to 10mV. In this case, the input signal is presumed to be preprocessed and amplified to 1 V for testing reasons. The waveforms of the suggested SAR ADC and its sub-circuits that emerged from simulation are displayed.

4.1 Sample and Hold Circuit

4.1.1 Pass-Transistor Switch

The V0_PLUS and V1_PLUS clock pulses are a train of 1V, 10MHz pulse trains. On the circuit shown Fig. 26, an input V2_PLUS of 1MHz and 2V_{P-P} is applied. A 1V supply voltage is used. The transmission gate arrangement without clock boosting circuit's sampled output I8_V_{out} is shown in Fig. 43. It is not possible to sample the low frequency input signal rail-to-rail using a 1V supply voltage and a 1V clock pulse. The amplitude distortion of the sampled signal is significant. This is as a result of the circuit's reduced clock signal and lower supply voltage.

The circuit holds the input at a specific level for the duration of the clock's negative cycle while tracking the input sinusoidal signal during the positive clock cycle. To maintain PMOS and NMOS devices in the active region simultaneously, an inverter, a hold capacitor, and a transmission gate-based switch are utilized.

The switch samples an analog input that is fed through it when it is in the "ON" position. The capacitor is subsequently charged. The capacitor stores and releases the sample value while the switch is in the "OFF" position. The control signal, sometimes referred to as the sampling frequency, in this instance needs to be higher than the input signal in order to produce an approximate representation of the input. This design also has the benefit of increasing the on-conductance between the input and output when the input is closer to the supply voltage.

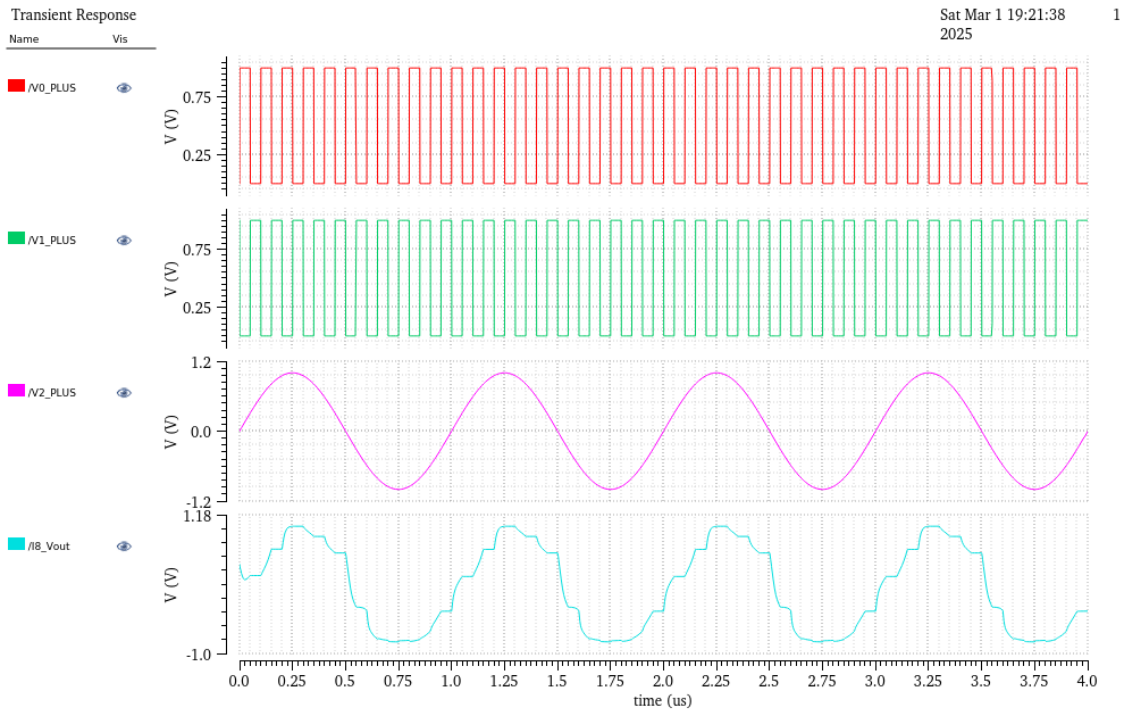


FIGURE 43. Transient analysis of pass-transistor switch.

4.1.2 S/H Circuit with Cascaded Simple Switches and Boosted Driver

The circuit is given an input signal with a frequency of $200Hz$ and an amplitude of 250 mV_{P-P}. A supply voltage of $1V$ is applied. A clock signal is a train of pulses with a frequency of $2KHz$ and an amplitude of $1V$. Fig. 44 display the output waveforms, respectively. Clock pulses with a $1V$ amplitude and a $1V$ power supply prevent rail-to-rail sampling [27] and [29]. These factors cause the sampled signal's amplitude to be significantly altered when using the clock signal and a low supply voltage (less than $1.2V$).

Fig. 44 shows the transient response of S/H circuit with cascaded simple switch. In this case, the booster driver's input is the same clock signal. According to simulation results, charge leakage has decreased and linearity has improved.

4.1.3 Proposed Sample and Hold Circuit

The input signal to the sampling circuit in Fig. 31 architecture is a sine wave with a frequency of $1MHz$. The circuit's operation is efficiently synchronized by the clock signals, which

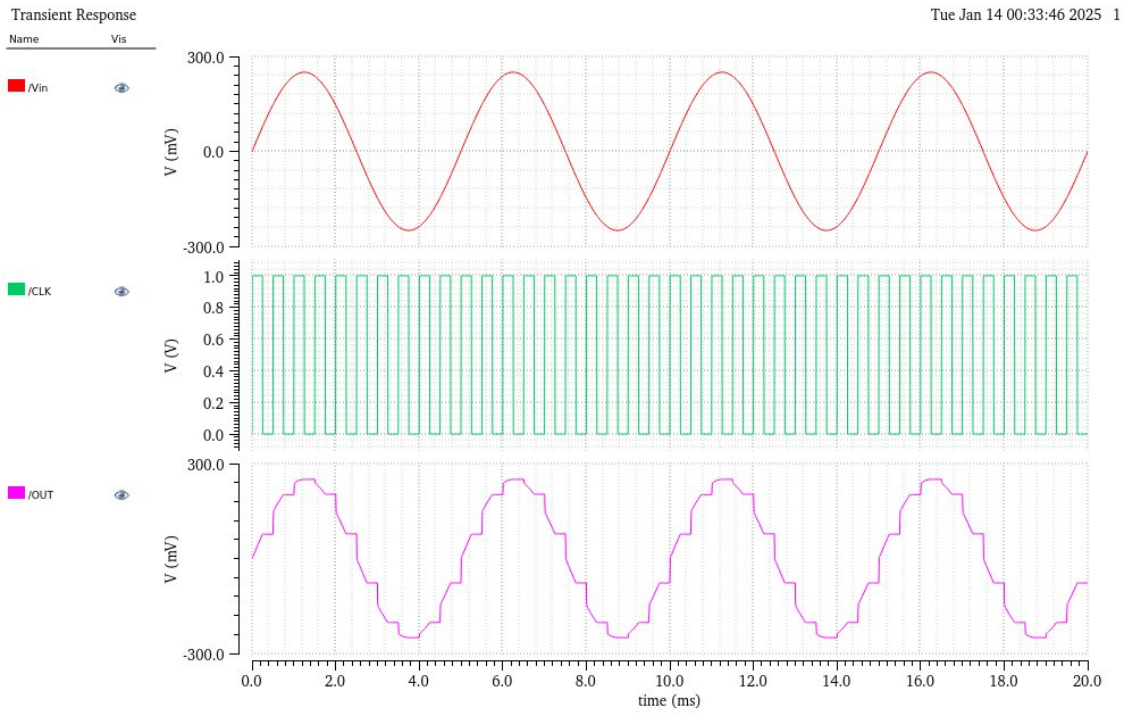


FIGURE 44. Transient analysis of cascaded simple switches and boosted drive.

control the sampling process by specifying the times at which the input signal is sampled.

The circuit is set up to record the instantaneous amplitude of the sine wave during the sampling phase when the clock pulse alternates between high and low states. Then, this amplitude is maintained constant so that other processing steps can function without interference from changes in the input signal. Maintaining signal integrity requires the ability to precisely sample and store the input signal, especially in biomedical applications where precision is critical.

This sampling procedure is demonstrated in Fig. 45, which also displays the relationship between the input sine wave, the switching behavior of the clock signal, and the output waveform. The S/H's output, shows that the sampled amplitudes of the sine wave are preserved during distinct time intervals. Each step of this output waveform, which approximates the input signal in a staircase fashion, represents the signal level that was recorded during the holding phase. The circuit guarantees that the signal's essential properties are maintained for precise digitization in the SAR ADC by successfully reducing signal loss or distortion during this operation. The general objective

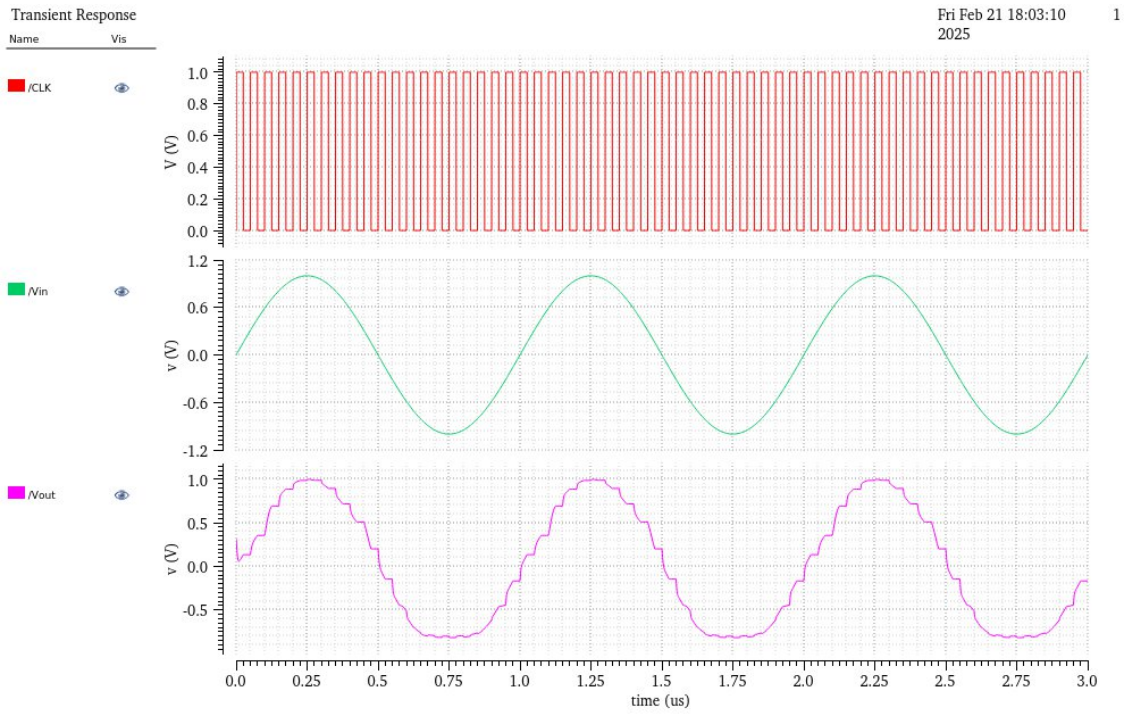


FIGURE 45. Transient analysis of proposed sample and hold circuit.

of the design effectively suppresses unwanted glitches during operation, enhancing signal integrity and ensuring a smoother conversion process.

The predicted power consumption of the various switching designs is listed in Table 4. The power usage in Watts shown in the table. The simulation results demonstrate that using a clock-boosting circuit lowers power consumption in addition to sampling distortion.

TABLE 4. Comparison of Different Sample and Hold Circuits

S.No.	Sample and Hold Circuits	Power Consumption (W)
1.	Transistor Gate Switch	1.60627p
2.	S/H circuit with cascaded simple switches and boosted driver	2.12648n
3.	Proposed Sample and Hold Circuit	31.6684p

4.2 Dual-Tail Comparator

A double tail comparator is a circuit that uses two tail MOSFETs to give the current two paths for discharge. Consequently, the circuit's delay is reduced. Comparing this construction to a

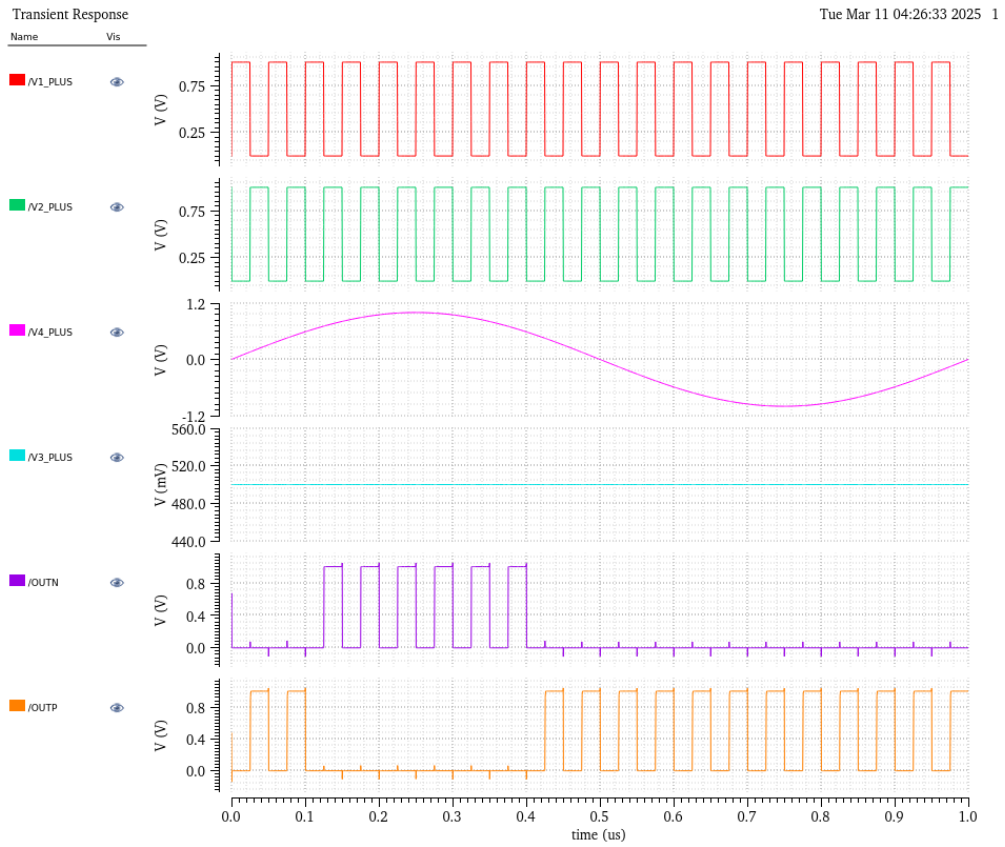


FIGURE 46. Transient analysis of double tail comparator.

standard comparator, there is also less transistor stacking.

The comparator (circuit seen in Fig. 32) response is displayed in Fig. 46. The comparator's positive leg receives one of the inputs, which is a sinusoidal signal /V2_PLUS with an amplitude ranging from 0 to $1V$ a frequency of the signal being $1MHz$. A steady DC signal with a value of $0.5V$ is provided as the reference input on the negative leg of the comparator. The output waveform displays the clock signals /V0_PLUS and /V1_PLUS, which are pulses with an amplitude of $1V$ and a period of $50ns$.

When the clock cycle is negative, the comparator outputs logic "0". Depending on whether the input is more or less than the reference voltage, the comparator compares the two inputs for the clock's positive cycle and outputs either logic "1" or logic "0".

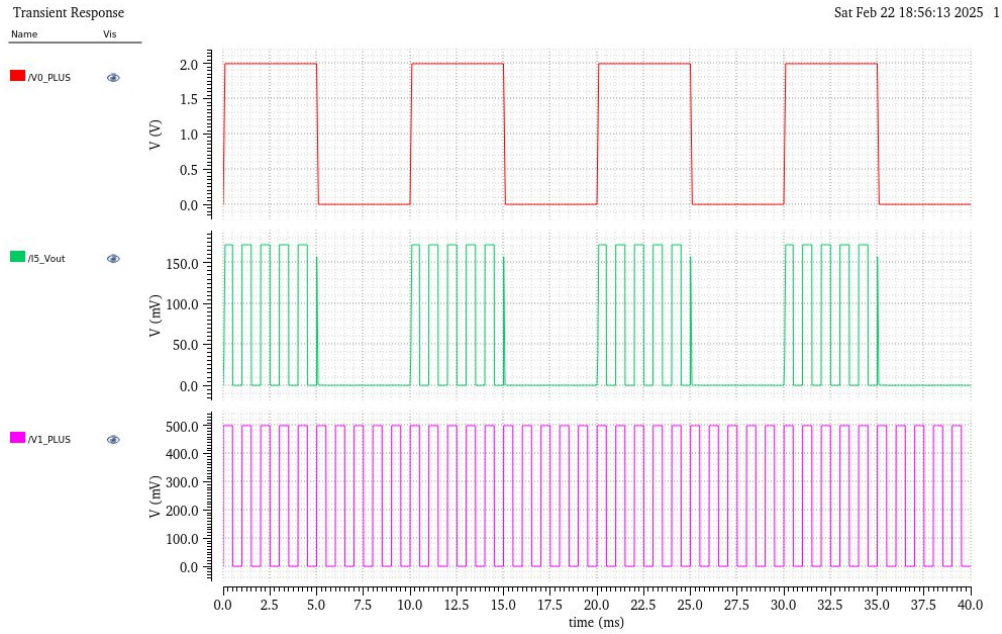


FIGURE 47. Transient analysis of 2:1 multiplexer.

4.3 Digital-to-Analog Converter

4.3.1 Split Capacitor Array DAC

A split array, 10-bit charge scaling DAC is built with the circuit's power consumption and chip space in mind. High DAC resolution is achieved by reducing the total area of the capacitor using the split approach in the charge scaling architecture. The SAR ADC uses a specially built 10-bit charge scaling DAC to redistribute charges.

The MUX response shown in Fig. 47 determines a specific voltage value based on the bit value of the input words, and the capacitor is charged accordingly. The MUX chooses the input, or the one connected to GND, if the applied input bit is LOW, and the voltage of the capacitor will be zero volts. In the event that the applied input bit is HIGH, the MUX chooses the input, which is connected to V_{ref} or the reference voltage, and charges the capacitor to the specified reference voltage.

The split capacitor arrayy DAC's response is depicted in Fig. 48 (Fig. 35 circuit design). Only pulse signals are accepted as input by the DAC. Ten pulses with different periods are supplied

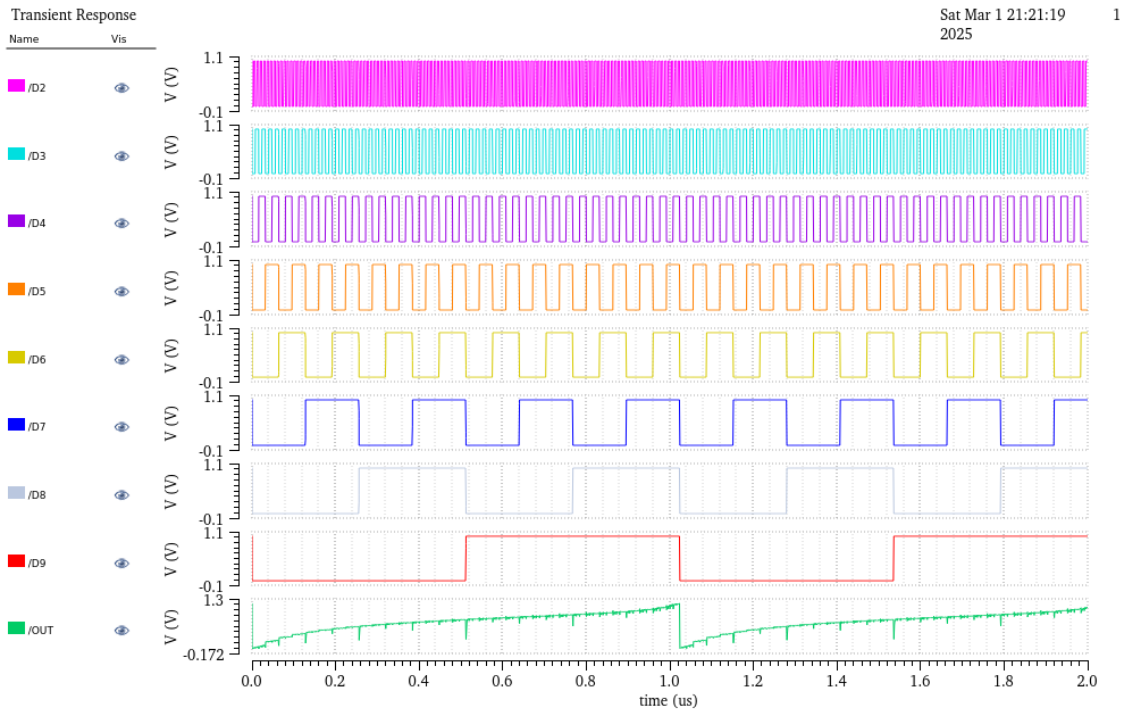


FIGURE 48. Transient analysis of split array capacitor DAC.

as input because this is a 10-bit DAC. Depending on how many of the pulse inputs are logic "1", the circuit produces an analog output. For instance, the DAC produces an analog signal with a value of $1.2V$ (V_{ref}) if every pulse that is supplied as an input to the circuit is a logic "1" (as also shown in Fig. 48). In DAC, where many capacitors are used to hold charge from the digital input and divide the analog signal into discrete values. This technique enhances linearity and resolution.

4.3.2 Binary Weighted DAC

With a resolution of 10 bits, the binary-weighted resistor DAC exhibits excellent output behavior with low power consumption and short settling time shown in Fig. 49. By using accurate resistor scaling in the sequence R, 2R, 4R,..., $2^{(N-1)}$R, the binary-weighted structure makes sure that the output voltage is linearly proportional to the applied digital input code. In the topology, the OPAMP serves as a buffer and summing node, providing high gain and guaranteeing quick settling to the intended output.

The CMOS switch is essential to this design because it minimizes voltage drops across the

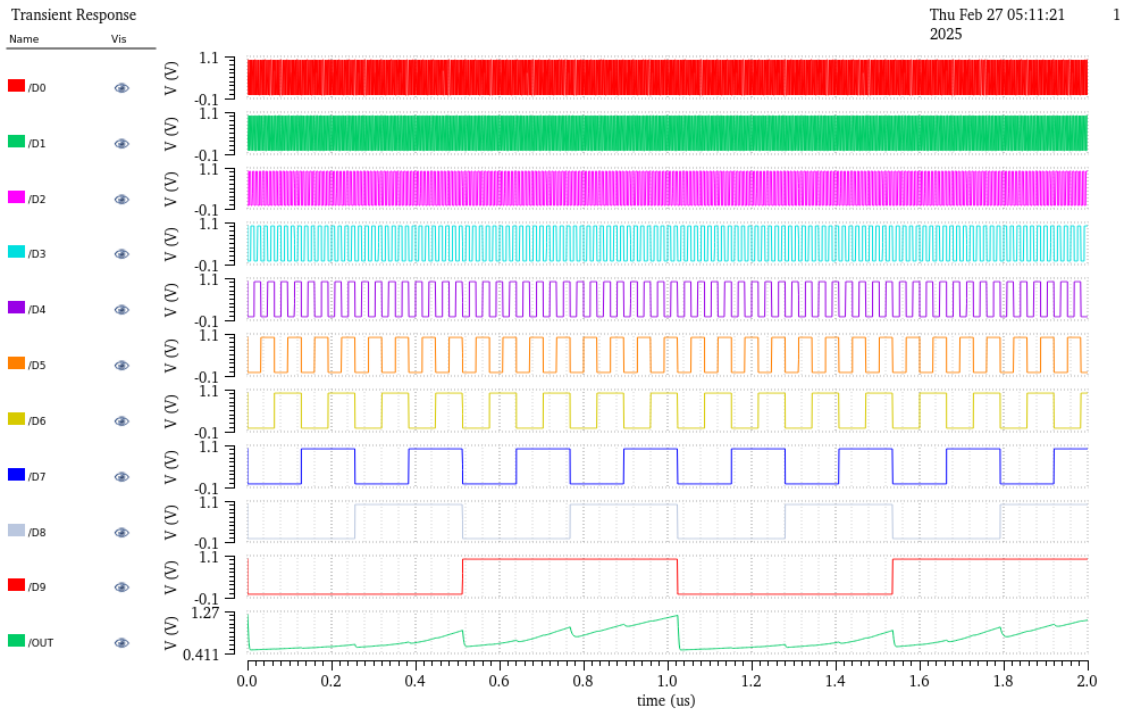


FIGURE 49. Transient analysis of binary weighted DAC.

switching parts by passing strong and stable logic levels with minimal on-resistance. The binary-weighted method is perfect for precision applications since it has faster output settling and much lower power consumption than alternative architectures like R-2R ladder DACs. The outcomes confirm that the planned DAC performs as intended, meeting the exacting specifications of biomedical applications where precise signal conversion and power efficiency are crucial.

This method guarantees excellent performance by careful resistor matching and layout design, which is crucial at a 10-bit resolution, in addition to achieving high-resolution and accurate DAC. Because of these benefits, a reliable and energy-efficient solution that works well in low-voltage, low-power applications is produced.

4.4 SAR Logic

The suggested SAR logic, as described in [19], produced the simulation results for the 10-bit SAR ADC. The transient response of the suggested SAR logic, which is based on the traditional SAR logic from Fig. 41 is depicted in Fig. 52. The simulation result of proposed D Logic is shown

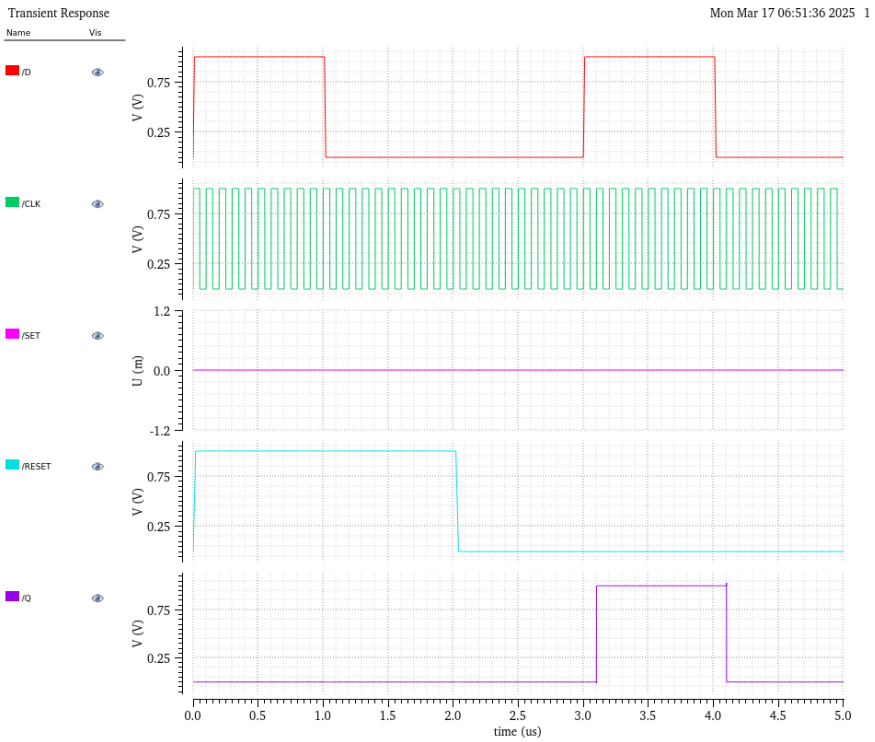


FIGURE 50. Transient analysis of proposed D logic.

in Fig. 50 and D Logic switch Fig. 51.

Three essential inputs are received by the SAR logic: the comparator input (COMP), the reset signal (RESET), and the clock signal (CLK). D9 through D0 are the ten pulses that make up the 10-bit SAR logic's output. Every time the clock signal has a positive edge, one of the output bits changes to logic "1". The value of the comparator's output determines whether the previously set output bit stays at logic "1" or changes to logic "0".

When the RESET signal is active, it sets the most significant bit flip-flop in the shift register and the outputs of the other flip-flops remain low. When RESET goes low, the ADC starts its conversion process. In the subsequent clock cycle, the MSB flip-flop is configured to logic '1', resulting in the generation of the corresponding DAC voltage. The counter subsequently transfers the logic '1' from the most significant bit to the least significant bit.

In each successive clock cycle, the output of the shift register determines the output of the

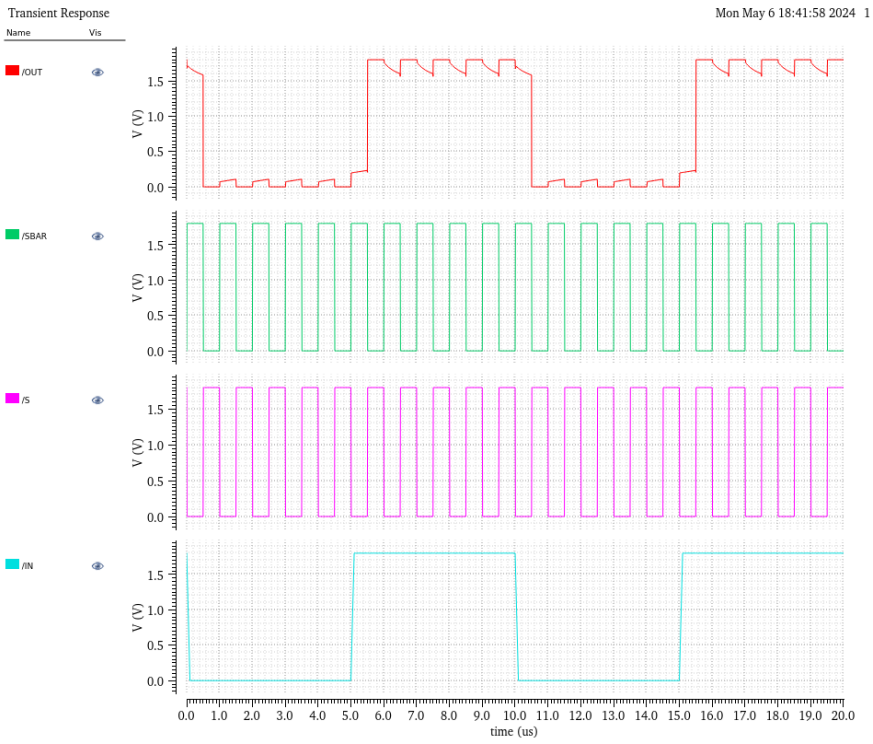


FIGURE 51. Transient analysis of D logic switch.

code register flip-flops. The outputs of the flip-flops, determined by the shift counter, function as the clock signal for the preceding flip-flop. On the positive edge of the clock, this flip-flop acquires the output from the comparator.

The comparator's output is used by the SAR control logic to compute the bit values in a sequential manner. A finite state machine controls the SAR's operation by sequentially driving the bit computation. If COMP is low, the bit is reset, and the next bit (D8) is tested. If COMP is high, the bit remains 1, and the next bit (D8) is also tested. This continues down to D0 (LSB). The results of the complete conversion are captured in the final cycle as shown in shown in Table 5.

4.5 SAR ADC

In the transient analysis of the proposed SAR ADC (circuit shown in Fig. 42), as depicted in Fig. 53 and 54, a sinusoidal input signal (/V_{in}) is utilized for testing purposes. We acquire this input during the negative phase of the sample clock signal. The sampled signal(/I23_V_{out}) is thereafter

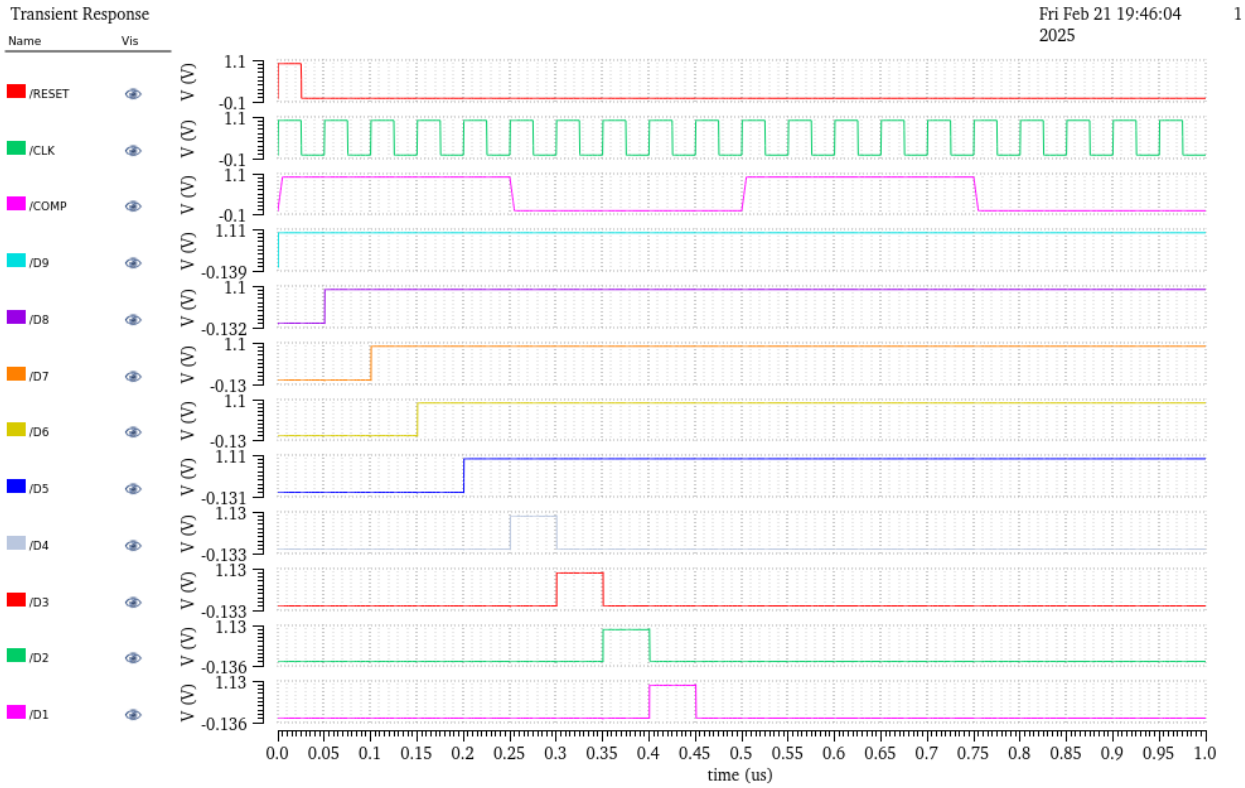


FIGURE 52. Transient analysis of proposed SAR logic.

TABLE 5. Finite State Machine

Cycles	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	COMP
0	1	0	0	0	0	0	0	0	0	0	A9=1
1	A9=1	1	0	0	0	0	0	0	0	0	A8=1
2	A9=1	A8=1	1	0	0	0	0	0	0	0	A7=1
3	A9=1	A8=1	A7=1	1	0	0	0	0	0	0	A6=1
4	A9=1	A8=1	A7=1	A6=1	1	0	0	0	0	0	A5=1
5	A9=1	A8=1	A7=1	A6=1	A5=1	1	0	0	0	0	A4=0
6	A9=1	A8=1	A7=1	A6=1	A5=1	A4=0	1	0	0	0	A3=0
7	A9=1	A8=1	A7=1	A6=1	A5=1	A4=0	A3=0	1	0	0	A2=0
8	A9=1	A8=1	A7=1	A6=1	A5=1	A4=0	A3=0	A2=0	1	0	A1=0
9	A9=1	A8=1	A7=1	A6=1	A5=1	A4=0	A3=0	A2=0	A1=0	1	A0=0
10	A9=1	A8=1	A7=1	A6=1	A5=1	A4=0	A3=0	A2=0	A1=0	A0=0	

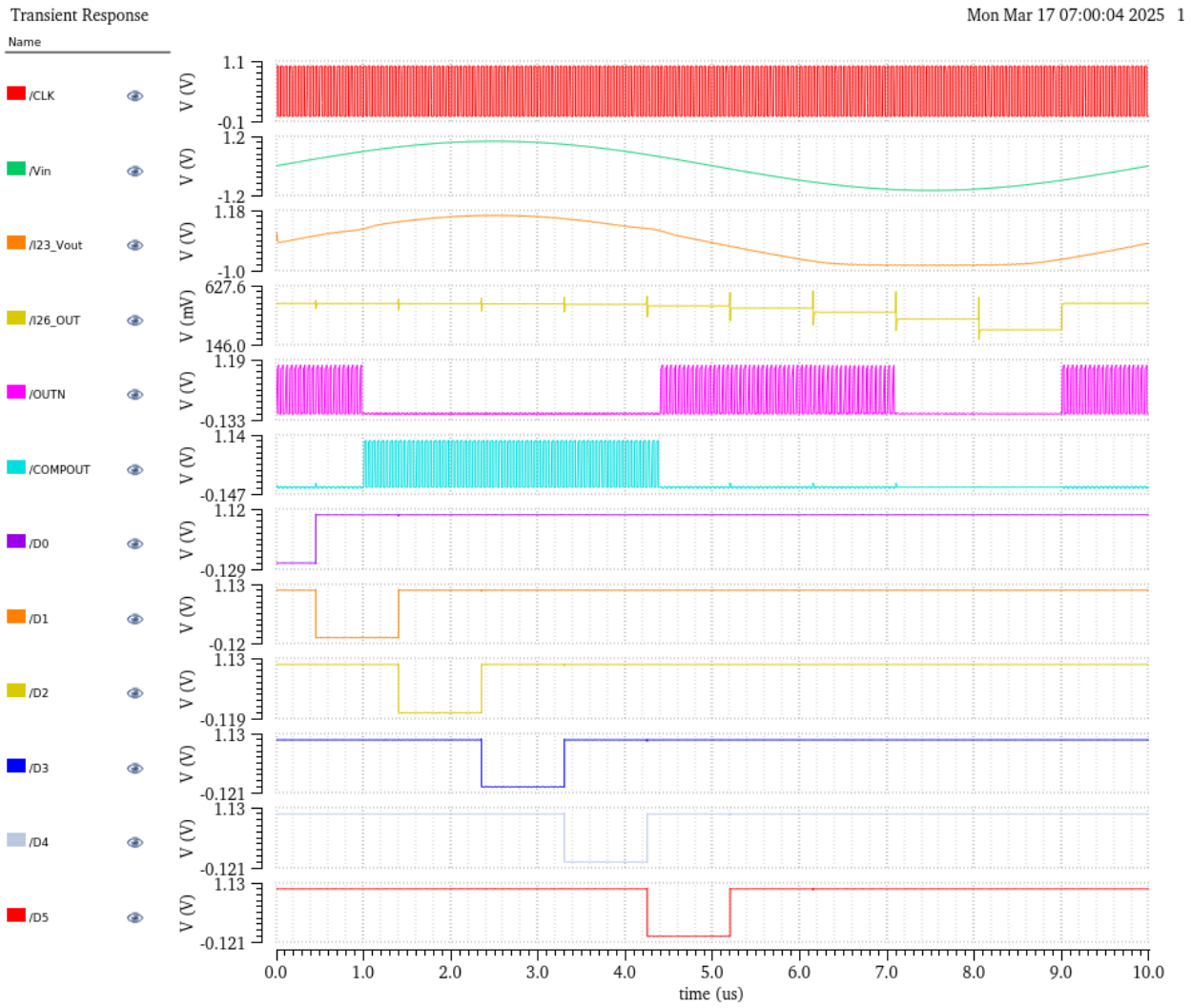


FIGURE 53. Transient analysis of proposed SAR ADC.

directed to the comparator, where it is compared with the output of the DAC(/I26_OUT). The comparator's output(/COMPOUT) is transmitted to the SAR logic, which produces a 10-bit pulse sequence. The pulse sequence(D0-D9) is input to the DAC, which subsequently generates an analog output(/I26_OUT). The analog output is routed back to the comparator, establishing a feedback loop, and the process perpetuates continuously. Table 6 contains the values of the parameters used in this design.

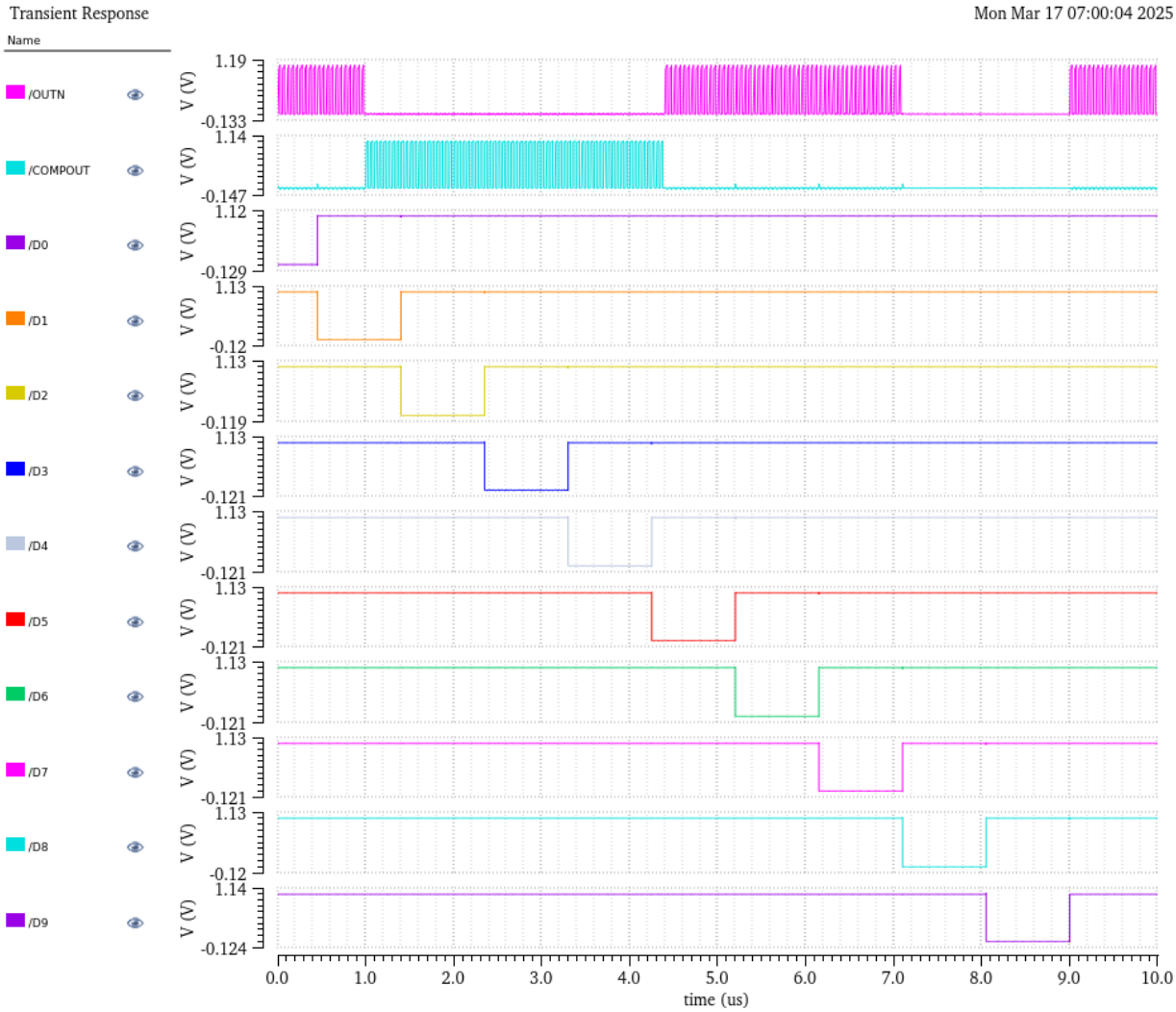


FIGURE 54. Output waveform for the proposed 10-bit SAR ADC.

TABLE 6. Parameters of the SAR ADC

Parameters	Values
V_{DC}	$1V$
No. of Bits	10
V_{sin}	$1V$, $0.1MHz$
Vpulse (Sample and Hold)	$1V$, Time Period=$50ns$
V_{ref}	$0.5V$
RESET	$1V$, Time Period = 1μs, Pulse Width=$25ns$

CHAPTER 5

LAYOUTS

This chapter emphasizes on the comprehensive techniques required for IC layout design, simulation, and verification. Layout design is a critical phase in circuit creation since it dictates the exact positioning and pathways of components. This activity ensures that the layout adheres to many production and functional criteria, enhancing circuit performance and spatial efficiency. Upon completion of the basic layout, we proceed to establish accurate connections between components to ensure signal integrity and adherence to timing specifications.

LVS and DRC are two crucial verification procedures that the design goes through when the layout is finished. The DRC ensures that the layout complies to manufacturing regulations, including minimum spacing, layer alignment, and width constraints. This process prevents errors during manufacture. LVS verification ensures that the physical layout precisely corresponds to the desired schematic. This ensures that all components and connections are accurately depicted for optimal system functionality.

Parasitic extraction is another important stage that involves analyzing unintentional resistances, capacitances, and inductances introduced during the layout process. These parasitic effects can profoundly affect circuit performance, especially at elevated frequencies, where they may diminish speed, gain, or stability. We incorporate these additional components using post-layout simulation to deliver an accurate depiction of the circuit's functioning. This method enables designers to implement the necessary modifications prior to the construction of the circuit. The approach utilizes tools such as Cadence Virtuoso, a multifaceted layout editor that interfaces effortlessly with other tools to facilitate the complete workflow, encompassing schematic design, layout, and verification. This chapter will examine each of these phases in detail, highlighting their significance in producing a practical, functional, and manufacturable integrated circuit design.

5.1 Layout Design and Planning

When designing the arrangement, a significant emphasis was placed on minimizing the effects of parasitic effects while simultaneously achieving high precision, low power consumption, and compactness. In order to put the idea into action, the 45*nm* CMOS technology was utilized. It was necessary to make use of sophisticated Electronic Design Automation technologies in order to guarantee that production specifications were strictly complied to and that layout-dependent effects were accurately represented. The Quantus system was used for parasitic extraction, and the Pegasus system was used for DRC and LVS work.

As part of this thesis, the comparator, which is one of the most important parts of the ADC, was made to work with little offset, fast speed, and low power consumption. During the layout phase, it was made sure that the comparator's parts, like the regeneration circuitry and differential pair transistors, were placed so that they were evenly spread out on both sides. This was done to reduce the number of mismatches that occur as a result of changes in the manufacturing process in which the components were constructed. Strategies like interdigitated transistor layouts and common-centroid placement were utilized, particularly for portions of the layout that were considered to be delicate.

The final layout of the comparator is depicted in Fig. 55, which highlights the compactness and symmetry of the design. To minimize the effects of parasitic capacitances and make certain that the main signal pathways were as brief as possible, extensive planning was done for the routing.

5.2 Design Rule Check and Layout Versus Schematic

To make sure the arrangement complied with the foundry's manufacturing specifications, it was put through a rigorous sequence of DRCs utilizing Pegasus. Every infraction, including minimum spacing, overlap, and width problems, was fixed gradually. Intermediate DRCs were carried out during the placement and routing stages to reduce errors in the early stages and guarantee that the layout could be built dependably while maintaining the circuit's intended functionality.

LVS verification was done after DRC to make sure the layout matched the original

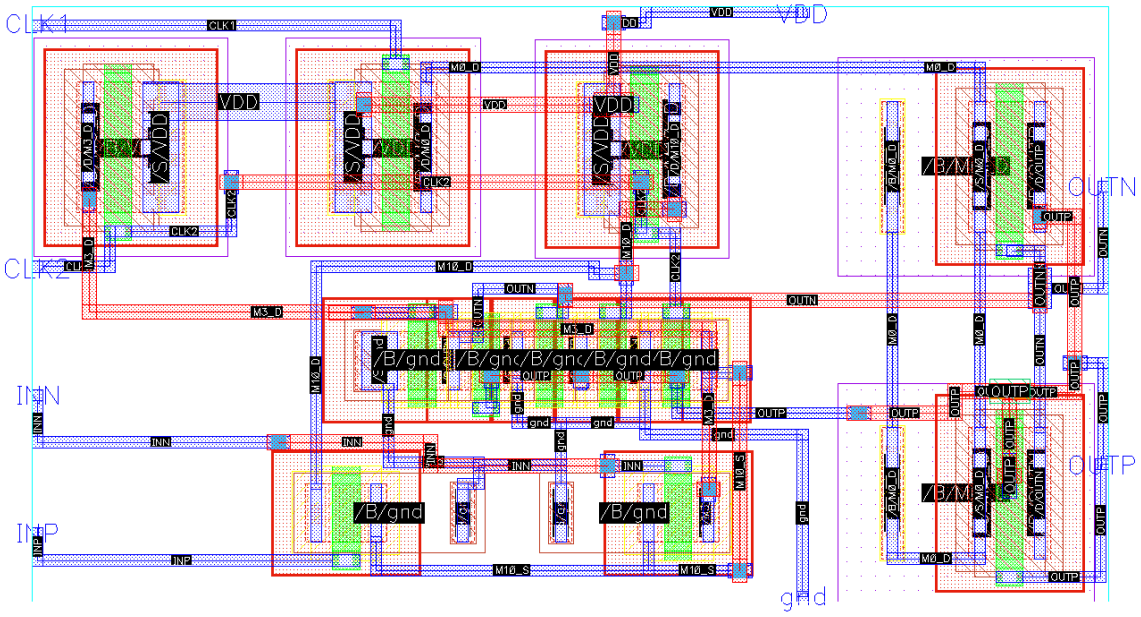


FIGURE 55. Comparator layout.

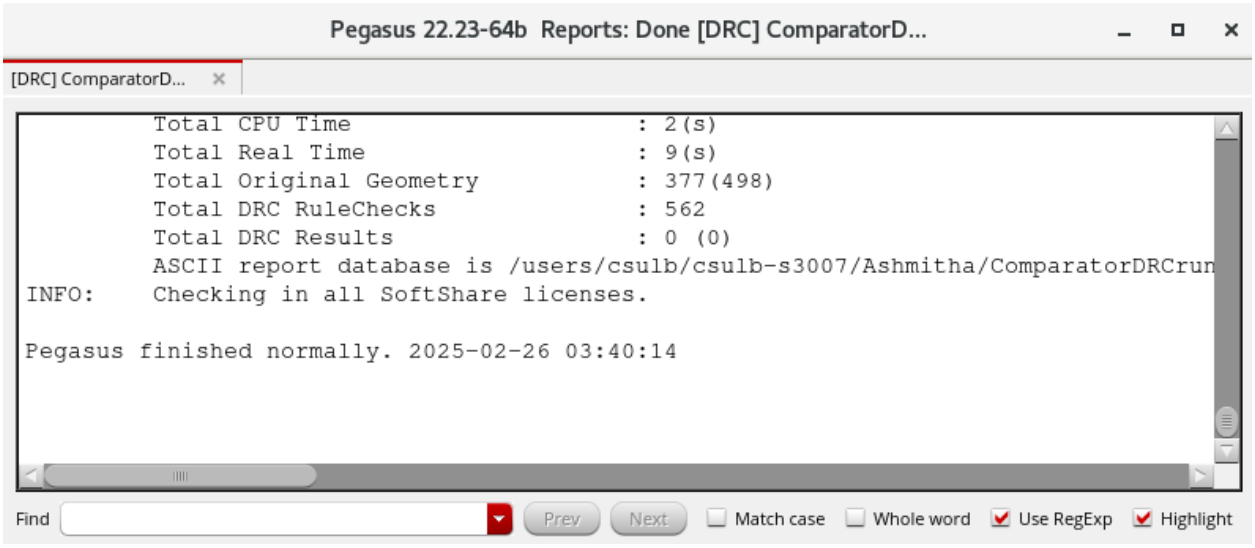


FIGURE 56. Design rule check.

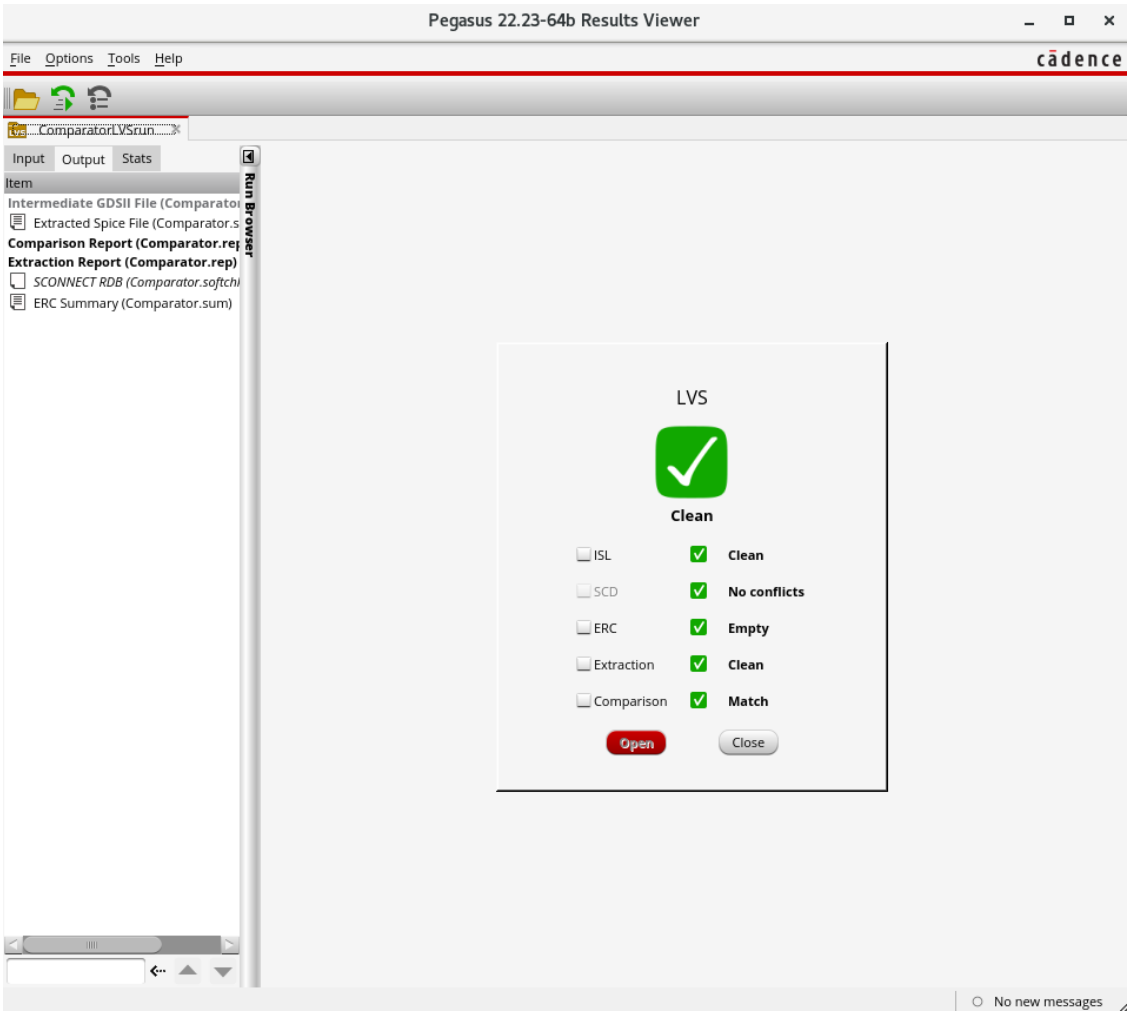


FIGURE 57. Layout versus schematic.

schematic. By using this step, the layout's accuracy in reflecting the intended circuit architecture was confirmed for each device and connection. The results of the DRC and LVS verification procedures are displayed in Fig. 56 and Fig. 57, respectively. These figures verify that the electrical design and physical execution are in line and show a neat, violation-free arrangement. If these verification processes are successful, the final layout will precisely implement the schematic functionality and be entirely compatible with process regulations.

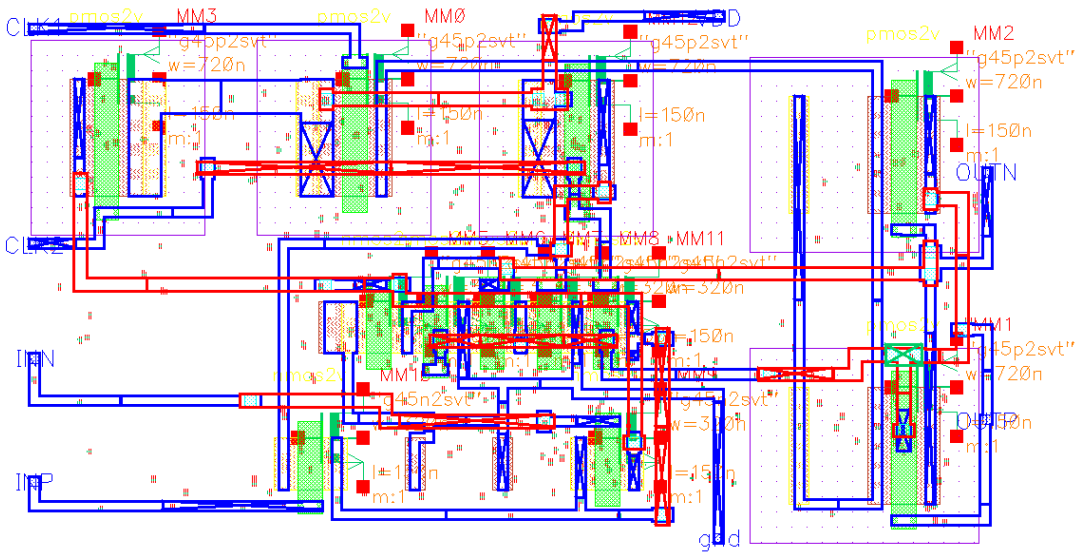


FIGURE 58. ev_extraction.

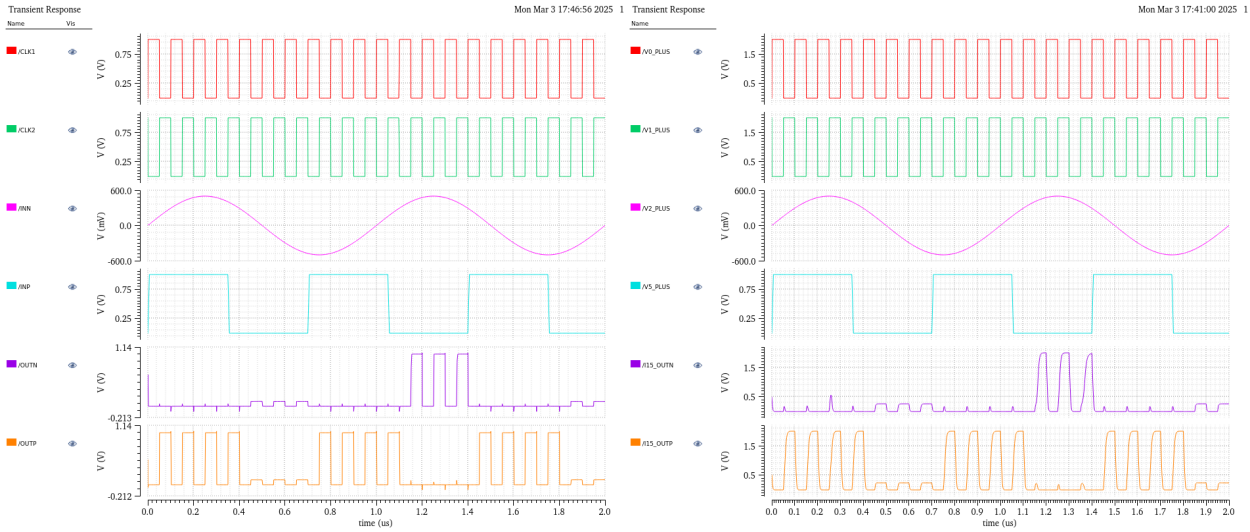
5.3 Parasitic Extraction with Quantus

Quantus was used for parasitic extraction, which involved extracting layout-dependent capacitances and resistances and integrating them into post-layout simulations. These parasitic elements significantly impact the ADC's performance, originating from closely linked circuit parts, routing routes, and vias. To keep speed and accuracy, extra care was taken to get rid of parasitics in the comparator's sensitive areas, like the input differential pair and the regeneration nodes.

We slightly modified the layout to address the manageable coupling capacitances and resistances between neighboring signal lines after eliminating the parasitics. The comparator layout's parasitic extraction is shown in Fig. 58, which also emphasizes the capacitance distribution and parasitic coupling across different nodes. For post-layout simulations, these parasitics were then modeled and incorporated into the circuit schematic. To prevent deteriorating the resolution and power efficiency of the ADC, quantitative analysis verified that the extracted parasitic capacitance and resistance stayed within acceptable bounds.

5.4 Post-Layout Simulations and Performance Verification

Post-layout simulations were conducted to verify the reliable operation of the SAR ADC. The simulation demonstrated that parasitic effects had no impact on the speed or accuracy of the ADC, hence confirming its reliability in low-power applications. The design performance consistency after incorporating parasitic effects was validated by comparing the simulation results before to and following the arrangement, as illustrated in Fig. 59. Parasitic capacitance resulted in a minor compromise in power and performance; still, layout optimization effectively mitigated this issue. Post-layout simulations demonstrated effective control of parasitic effects, ensuring that the ADC maintained its superior performance characteristics, including compact dimensions and low power consumption.



(a) Pre-layout simulation. **(b) Post-layout simulation.**

FIGURE 59. Simulations with parasitic effects.

The design and verification of the SAR ADC layout were effectively executed utilizing sophisticated tools and procedures. The comparator layout was carefully designed to achieve a balance between performance, area efficiency, and manufacturability. The ADC configuration was subjected to DRC and LVS verifications to guarantee adherence to process design regulations and precise device operation. Post-layout simulations verified that parasitic effects were successfully

managed. These findings indicate that the proposed architecture is ideal for biological applications requiring compact integration, minimal power consumption, and high precision.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

This work details the implementation of a 10-bit SAR ADC functioning at $20MS/s$ with a supply voltage of $1V$, fabricated using a $45nm$ CMOS process. The ADC uses a S/H replacement, a split array DAC along with a binary weighted DAC, a double-tail comparator, and new SAR control logic that adds an extra logic element to SAR logic code registers instead of the traditional DFF. The double-tail comparator demonstrates a reduction in power dissipation when compared to the traditional dynamic comparator. A DFF is incorporated into the output of the comparator to latch the comparator's output prior to its transmission to the SAR Logic.

6.2 Future Scope

This work presents the implementation of a 10-bit SAR ADC for biomedical applications. This project will be extended to include on-chip layout design and fabrication of the ADC components, thereby facilitating practical hardware testing and evaluation. The entire performance will be enhanced, while the system's power consumption and dimensions will be reduced through the total integration of all components—S/H, DAC, comparator, and SAR logic—onto a single chip, making it suitable for portable biomedical devices.

This solution could also be used to lower technological nodes (like $28nm$, $22nm$, or even $16nm$) to leverage enhanced scalability, potentially yielding advantages such as reduced power consumption, minimized area, and improved performance. The shrinking of the process node enables the incorporation of additional functionalities on the chip, enhancing the SAR ADC as a more robust component for complex biomedical systems.

In the future, the proposed model may enhance its capabilities to achieve ideal low-noise and high-speed performance, which is essential for precise signal processing in biomedical applications. The expansion of the architecture for the simultaneous collecting of data from several biomedical sensors significantly enhances the management of multi-channel systems, hence

significantly boosting the adaptability of the ADC within medical devices.

With the increasing demand for precise medical measures, this project may extend beyond 10 bits to higher resolutions such as 12, 14, or 16 bits. The enhanced resolution it provides would assist applications requiring precise measurements, such as EEG, ECG, and other critical biomedical functions.

Incorporating this solution with certain wireless communication functionalities could facilitate real-time data transmission to cloud-based monitoring and analysis systems. This may facilitate continuous remote monitoring of patients health, allowing timely diagnosis and intervention.

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